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Professor Figueroa-López

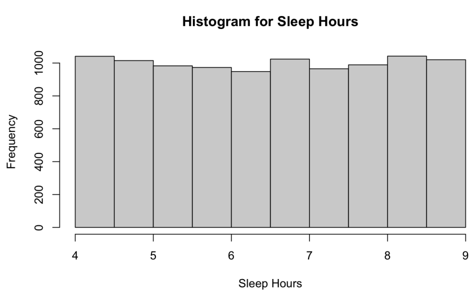
Linear Statistical Models

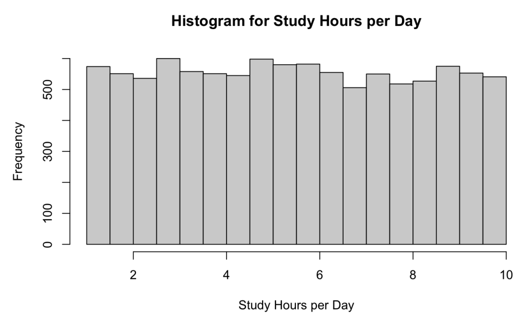
27 April 2026

LSM Final Project: Student Performance Analytics

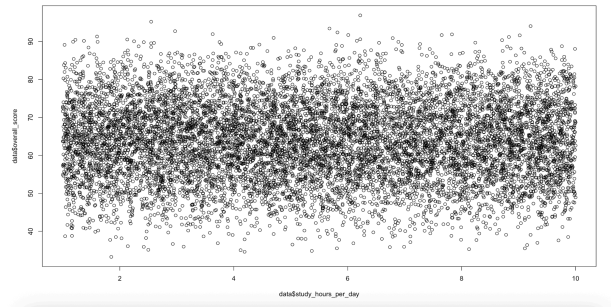
This project uses the Student Performance Analytics Dataset that can be found on Kaggle under the MIT license. The dataset contains detailed records of student academic performance along with behavioral and background factors. Each instance in the dataset represents an individual student. The response we are aiming to model is the Overall Score, which is a continuous variable. We will run a series of statistical tests on the dataset to determine if it can be modeled by a linear regression.

The dataset is observational with 10,000 instances of students with information about their study habits, attendance, assessment scores, and personal learning conditions. There are no nonresponse nor missing values. The categorical variables in the dataset are Gender, Internet Access, Extra Classes, and Parent Education. For the purposes of analysis, Gender, Internet Access, and Extra Classes are encoded as 0 or 1. Parent Education is ordinally encoded with Bachelor's Degree as 0, Master's degree as 1, and PhD as 2. The figures below show the histograms for some key categorical variables. As can be seen these variables are fairly uniform, with minor peaks.





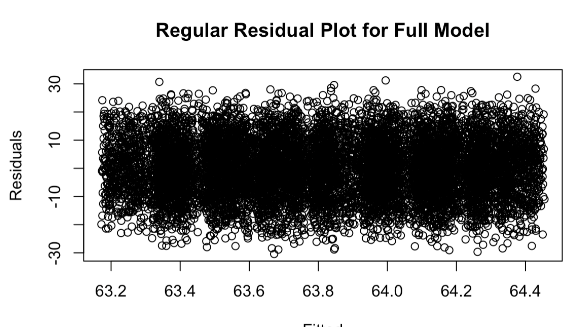
The numerical variables we are using in the regression are Study Hours per Day, Attendance Percentage, and Sleep Hours. As can be seen through the figure below, it is difficult to see a trend in the relationship between the variables, possibly due to the size of the dataset.

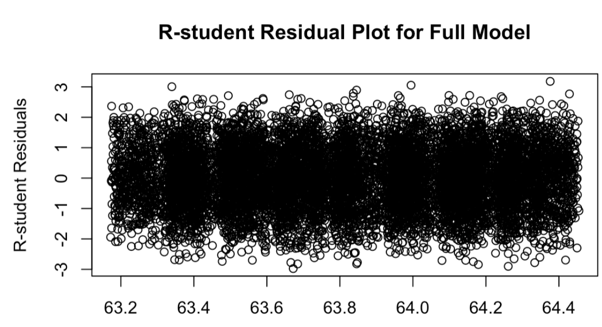


We made the decision to remove the feature Student ID since it is unique for each student and it does not convey any additional information. The Grade feature is a categorical version of the overall score, so it will also be removed from the dataset as well. Initially when we ran a linear regression on the data, it returned a model with an R^2 value of 1. This is due to data leakage. The overall score can be calculated by summing a percentage of the Assignment Score, Midterm Score, Final Exam Score, and Participation Score. In order for the problem to be more interesting, we will remove these features to see how student habits and backgrounds will impact their final score.

The dataset we will use for the remainder of the project has features Gender, Study Hours per Day, Internet Access, Extra Classes, Parent Education, and Sleep Hours. The model has an R^2 value of 0.0233 and adjusted R^2 of 0.0224. The regression is significant since the p-value is below 2.2e-16. Attendance Percentage and Internet Access are significant features. Looking at the signs of the feature estimates, it is not intuitive that Study Hours per Day is negative. One could justify that smarter students study less than those that are struggling. Negative sleep hours may be justified by the fact that smarter students sleep less since they are working more. Lastly, negative Parent Education features may be due to the fact that students with highly educated parents may have a lack of motivation to do well.

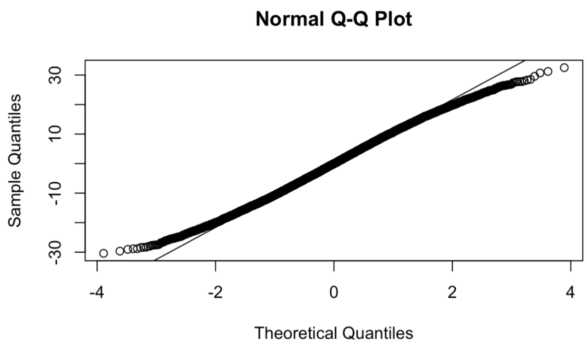
To complete the model diagnostics for our full model, we checked the constant variance and normality assumptions, and the presence of large leverage points, outliers in y, and influential points. First, to check the constant variance assumption, we plotted the fitted values against both the residuals and the R-student residuals:

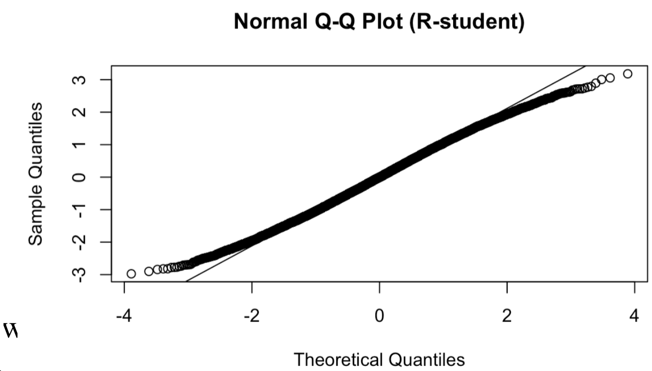




In both residual plots for the full model, the points are spread out relatively evenly, with no clear pattern. Therefore, the constant variance assumption does not seem to be violated.

Next, we checked the normality assumption using qq-plots and normality tests. The qq-plots are as follows:





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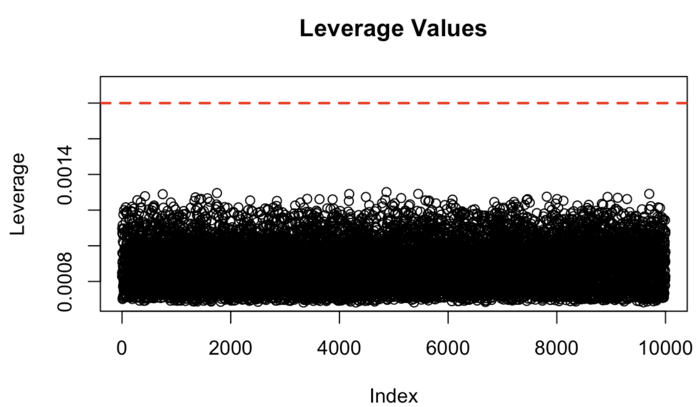
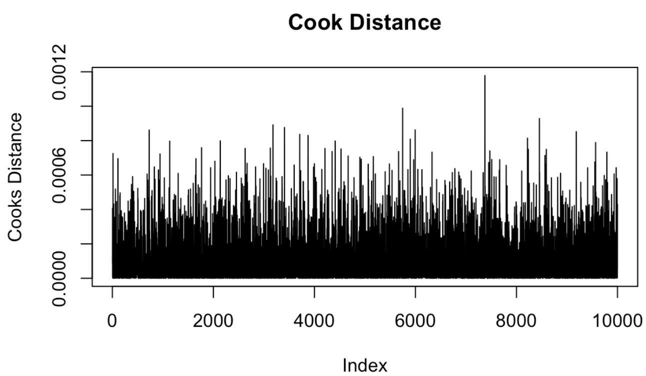
ends. From these plots alone, we can conclude that the normality assumption is not violated. In addition, we decided to use the Kolmogorov-Smirnov test to formally test the normality assumption. We used this test instead of the Shapiro test because we have a large sample size greater than 5000. Based on the Kolmogorov-Smirnov test, we rejected normality with a p-value of 0.0323; however, with a large sample size, even mild deviations from normality may be detected. So, by the Central Limit Theorem, because of the large sample size, the sampling distributions of the regression coefficients are approximately normal. Therefore, we conclude that the normality assumption is satisfied.

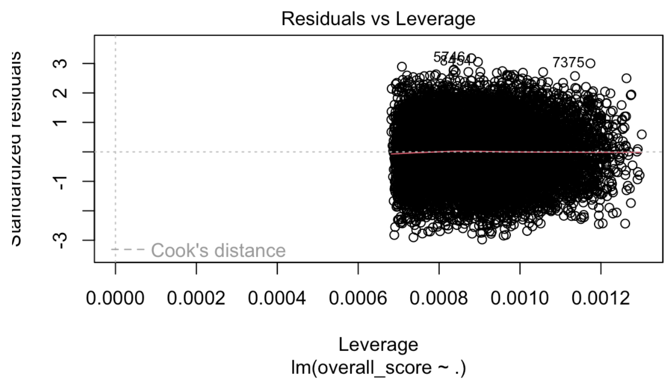
To check for large leverage points and outliers, we plotted the leverage values and calculated Bonferroni's critical value. In the plot of the leverage values,



no leverage point exceeds the limit of 2*p/n, where p represents the number of coefficients in our model (including the intercept), and n represents the sample size. Therefore, we conclude there are no large leverage points. To find any outlier points in y, we found Bonferroni's critical value to be -4.567283, and the maximum R-student residual to be 3.180333. Since 3.180333 < 4.567283, we conclude that there are no outlier points in y.

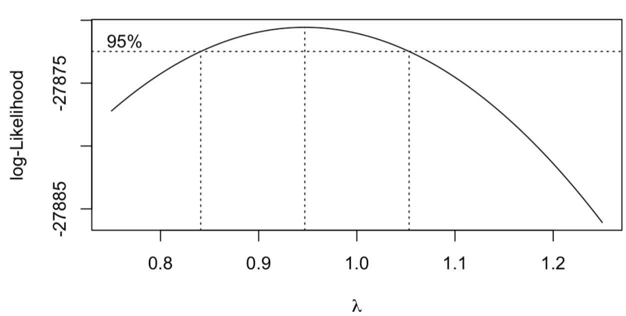
Finally, we checked for influential points using Cook's Distance and by plotting the residuals vs. leverages:





We found the maximum Cook's Distance to be below the critical threshold, so there are no influential points.

Although the errors for our model were all reasonably homoskedastic, we decided to try a transformation using the Box-Cox Transformation:



From the plot above, we see that 1.0 falls within the confidence interval for lambda. Therefore, we conclude there is no good reason to transform the response variable and the original model is appropriate.

The next step was for us to consider the possibility of autocorrelation in our data. Although we had earlier removed the Student ID, for the sake of this task, we added it back into the data we were analyzing. If the dataset had some sort of pattern based on the students' IDs, we would see serial correlation, causing issues for our model. Upon conducting a Durbin-Watson test, however, we would see that this is not a concern for our dataset. The test produced a value of 2.0089 with a p-value of 0.6677, suggesting that there is no significant autocorrelation in our data, as the DW value was nearly 2 and the p-value was above $\alpha = .05$. Although Student IDs would not be considered in our model regardless, it is validating to find that it has zero predictive power.

We also must account for the possibility of heteroscedasticity. Our OLS would be inefficient in the case that the variance depends on the value of an independent variable. To test this, we applied the Breusch-Pagan test to our model. This produced the ideal result of a test statistic of 2.2566 with 9 degrees of freedom and with a p-value of 0.9867. This suggests that there is no heteroscedasticity in our model. Despite this, we created a WLS estimator just in case. For our WLS, we considered that Study Hours per Day might impact the variance of a student's overall score. We sorted the data by Study Hours per Day and split it into 20 quantiles. For each group, we calculated the mean Study Hours per Day and the variance in Overall Score. When we ran a regression based on modeling the variance as a function of the means. We then fed our original data back into this variance model to predict the specific variance for every single observation in our dataset. Finally, to create the actual weights for the WLS model, we took the reciprocals of these predicted variances. With these weights, we created our WLS model. As expected, this weighted model performed little better than the original least squares model. The adjusted R^2 of this model was 0.02255, meaning it only increases by 0.00031 from the original model. This makes sense considering that our Breusch-Pagan test had a p-value of 0.9867. Evidently, our data does not suffer from heteroscedasticity. Because the variance was already perfectly constant, applying weights to force a correction was redundant.

In addition to analyzing model diagnostics, we also needed to test for the potential of multicollinearity within the full model. To assess this, we used three different approaches by analyzing pairwise correlations, variance inflation factors (VIF), and condition numbers.

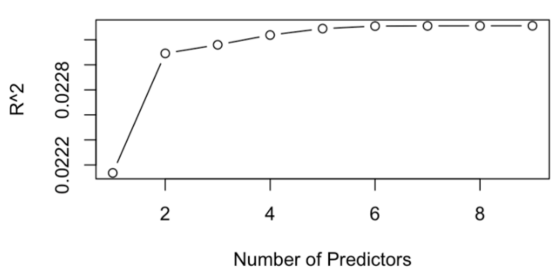
First, looking into the pairwise correlations between all of the quantitative covariates, we saw that all of the correlations are very close to zero, with the maximum value being around

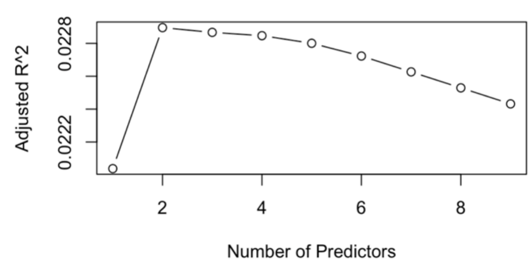
0.149 that exists between attendance percentage and overall score. With this information, we can conclude that the quantitative variables have little to no linear relationship amongst each other providing at least one sign of proof that multicollinearity is not present.

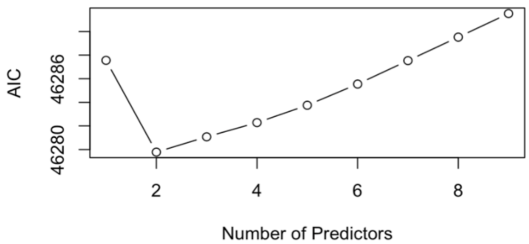
Second, we needed to measure VIF to assess if any one predictor's estimates are being distorted due to them being potentially correlated with other predictors. To do this, we applied the 'vif()' function in R to our full model. This returns the generalized VIF, degrees of freedom, and most importantly for this analysis, the adjusted VIF. Upon observing the output for the function, it was apparent that all of the adjusted VIF values for each predictor was essentially ~ 1.000. This result is an indication that the coefficient estimates are very stable which leads us with more evidence that there is no multicollinearity.

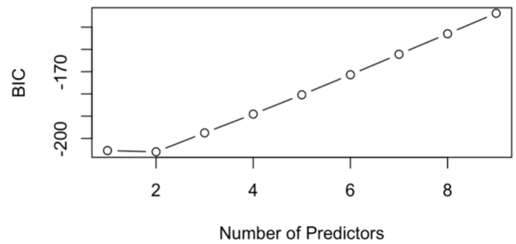
Finally, condition numbers are needed to further confirm that there is no evidence of multicollinearity. Due to the scaling differences and dummy variable encoding of categorical variables, standardization was required to assess solely for multicollinearity. After scaling the data, condition numbers were derived from eigenvalues of the product of the standardized model and the transpose of the standardized model. The resultant condition indices ranged from 1.000 to 1.996 which again, suggest there is no evidence of multicollinearity.

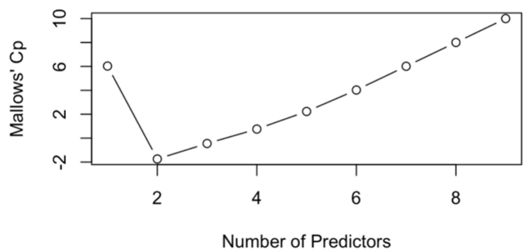
Due to the number of predictors in this dataset being relatively small, an exhaustive search is needed to determine the optimal subset of predictors for modeling the overall score. We evaluated this based on R^2, adjusted R^2, AIC, BIC, and Mallows' Cp. As expected, and shown from the graphs below, R^2 increases monotonically with the number of predictors, as the optimal model chosen contains 9 variables, which is representative of the full model. When looking at the graphs for R^2 adjusted, AIC, BIC, and Mallows' Cp, we saw that the optimal number of predictors when choosing a suitable reduced model was set at 2 predictors, indicating that adding any of the other predictors does not meaningfully improve model performance.







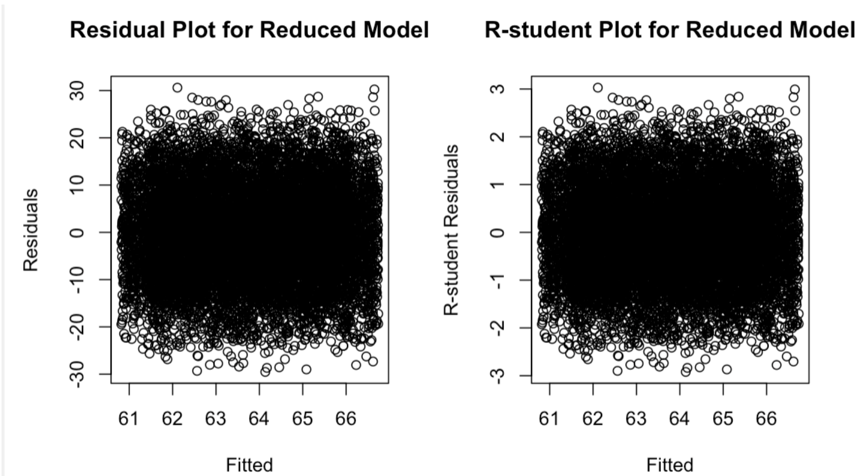




Overall, the selected model according to all metrics, apart from R^2, includes the variables ‘attendance_percentage’ and ‘internet_accessYes’. The resultant formula of the model is given as follows: overall_score = 57.287 + 0.0882(attendance_percentage) + 0.6325(internet_accessYes). Both predictors seem to share a positive relationship with the response variable, which indicates that higher attendance and having access to the internet are associated with overall performance. Overall, the agreement between the different criteria (R^2 Adjusted, AIC, BIC, and Mallows’ Cp), in selecting the same two-predictor model provides strong evidence that it is the preferred reduced model for predicting the response variable.

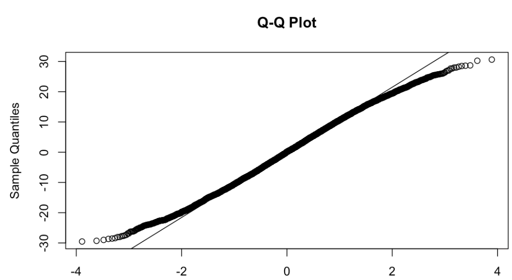
Next, we applied backward and forward stepwise model selection methods. According to Topic 18, these methods iteratively add or remove predictors based on the chosen selection criterion, which in our case is the AIC. AIC measures model fit and complexity, and penalizes models that have too many variables. This step function in R chooses the model that minimizes the AIC at each step. After running both the backward and forward methods, we found that they selected the same reduced model. This model has an AIC of 75660.55, which is lower than the full model’s AIC of 74672.29. This implies the reduced model has a better tradeoff between fit and complexity than the full model. Even though we started with many more predictors, the model selection showed us that only attendance and internet access were necessary to explain student performance. Therefore, we chose this model to proceed with for further analysis.

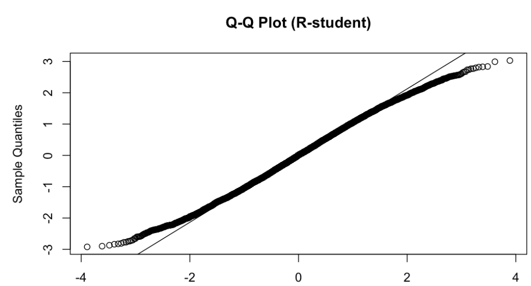
After selecting this model, we re-ran the model diagnostics that we used for the full model. This included checking constant variance and normality assumptions, and the presence of large leverage points, outliers in y, and influential points. First, looking at constant variance, we looked at both the residual plot and the R-student residual plot. These can be seen below:



In both plots, the residuals are randomly scattered with a relatively constant spread. We also see that there's no visible funnel pattern. This tells us that the constant variance assumption is satisfied for the model.

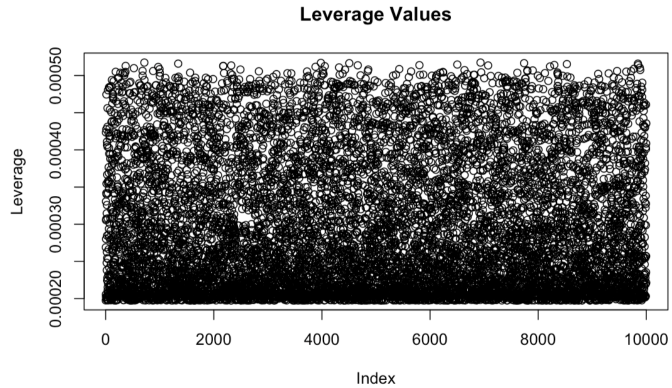
Next, checking normality assumptions, we looked at Q-Q plots and a KS test. The Q-Q plots can be seen below:





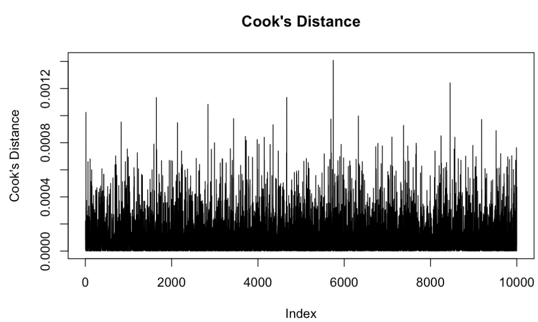
The Q-Q plots show that most points lie close to the line, with only slight deviations in the tails. The KS test returned a p-value of 0.008656, which rejects normality. However, given that we are dealing with such a large sample size, we would expect some small deviations. So, we still consider the normality assumption to be reasonable.

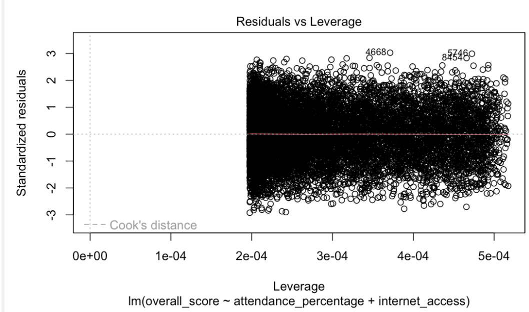
Next, we checked for the presence of large leverage points or outliers. To do this, we plotted the leverage values, which can be seen below:



The maximum leverage value was 0.000517, which is below the cutoff of 2p/n. We have 3 parameters and 10,000 data points, so our cutoff is 0.0006. Since there are no observations greater than this cutoff, we can safely conclude that there are no large leverage points. To check for any outliers, we calculated Bonferroni's critical value, which was 3.028. The largest absolute value of the R-student residuals was 4.567, which exceeds the critical value. This tells us that there is at least one potential outlier.

Finally, to identify any influential observations, we looked at Cook's distance and plotted the residuals vs. leverages. Our outputs can be seen below:





The maximum Cook's distance was very small at approximately 0.00141. The plot of residuals vs. leverages also did not show any standout points. Considering both of these results together, we concluded that there were no influential observations affecting the model. Overall, the reduced model satisfied most regression assumptions. There were slight deviations from normality and the presence of a few small outliers, but no major issues that could jeopardize the validity of our model.

To conclude, we evaluated multiple regression models to better understand which factors influenced student performance. We started with a full model, but ultimately found a reduced model could explain the variation in student performance. The simpler model highlighted that attendance percentage and internet access were the most important predictors. This implies that student engagement and access to resources are crucial for academic success. We then validated this model with diagnostic tests, showing us that our model was statistically sound while still remaining easy to interpret.

Work Cited

1. https://www.kaggle.com/datasets/borovai0/student-performance-analytics-dataset

SDS 4130 Final Project

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2026-04-24

Task 1

First we need to understand the purpose of the study and how the data was collected. The dataset details records of academic performance along with behavioral and background factors. Below is the first six instances of the dataset and the numerical summaries.

```
original_data <- read.csv("student_performance_data.csv")
#https://www.kaggle.com/datasets/borovai0/student-performance-analytics-dataset
(head(original_data))
```

```
##   student_id gender study_hours_per_day attendance_percentage assignment_score
## 1     100000   Male                4.54                 69.98            36.47
## 2     100001 Female                5.26                 84.80            34.25
## 3     100002   Male                8.69                 73.76            72.29
## 4     100003   Male                4.06                 45.00            97.63
## 5     100004   Male                8.83                 51.13            65.19
## 6     100005 Female                1.79                 53.16            33.61
##   midterm_score final_exam_score participation_score internet_access
## 1         70.70            53.10               17.96             Yes
## 2         27.92            87.17               11.29              No
## 3         70.92            99.61               76.10              No
## 4         31.73            88.85               33.55              No
## 5         78.28            54.23               88.99              No
## 6         54.26            79.79               58.23              No
##   extra_classes parent_education sleep_hours overall_score grade
## 1            No           Master        8.09       52.3480     D
## 2           Yes         Bachelor        4.73       53.9485     D
## 3           Yes              PhD        8.73       82.0375     B
## 4            No         Bachelor        8.22       66.4110     C
## 5            No         Bachelor        8.59       65.6005     C
## 6            No         Bachelor        5.43       59.3525     C
```

```
(summary(original_data))
```

```
##    student_id        gender          study_hours_per_day attendance_percentage
##  Min.   :100000   Length:10000       Min.   : 1.000      Min.   : 40.00
##  1st Qu.:102500   Class :character   1st Qu.: 3.220      1st Qu.: 55.64
##  Median :105000   Mode  :character   Median : 5.420      Median : 70.78
##  Mean   :105000                      Mean   : 5.468      Mean   : 70.49
##  3rd Qu.:107499                      3rd Qu.: 7.740      3rd Qu.: 85.20
##  Max.   :109999                      Max.   :10.000      Max.   :100.00
##  assignment_score midterm_score    final_exam_score participation_score
##  Min.   :30.00    Min.   : 25.00   Min.   : 30.01   Min.   :10.00
##  1st Qu.:47.32    1st Qu.: 43.52   1st Qu.: 47.40   1st Qu.:32.95
##  Median :64.77    Median : 62.05   Median : 65.21   Median :55.05
##  Mean   :64.72    Mean   : 62.32   Mean   : 64.98   Mean   :55.10
##  3rd Qu.:81.76    3rd Qu.: 81.23   3rd Qu.: 82.50   3rd Qu.:77.76
##  Max.   :99.99    Max.   :100.00   Max.   :100.00   Max.   :99.99
##  internet_access    extra_classes      parent_education    sleep_hours
##  Length:10000       Length:10000       Length:10000       Min.   :4.000
##  Class :character   Class :character   Class :character   1st Qu.:5.220
##  Mode  :character   Mode  :character   Mode  :character   Median :6.520
##                                                           Mean   :6.505
##                                                           3rd Qu.:7.793
##                                                           Max.   :9.000
##  overall_score      grade
##  Min.   :33.26   Length:10000
##  1st Qu.:56.53   Class :character
##  Median :63.78   Mode  :character
##  Mean   :63.83
##  3rd Qu.:71.16
##  Max.   :96.88
```

```
(nrow(original_data))
```

```
## [1] 10000
```

```
data <- subset(original_data, select = -c(student_id, grade))
```

```
data$gender <- factor(data$gender)
summary(data$gender)
```

```
## Female   Male
##   4987   5013
```

```
data$internet_access <- factor(data$internet_access)
summary(data$internet_access)
```

```
##   No  Yes
## 4939 5061
```

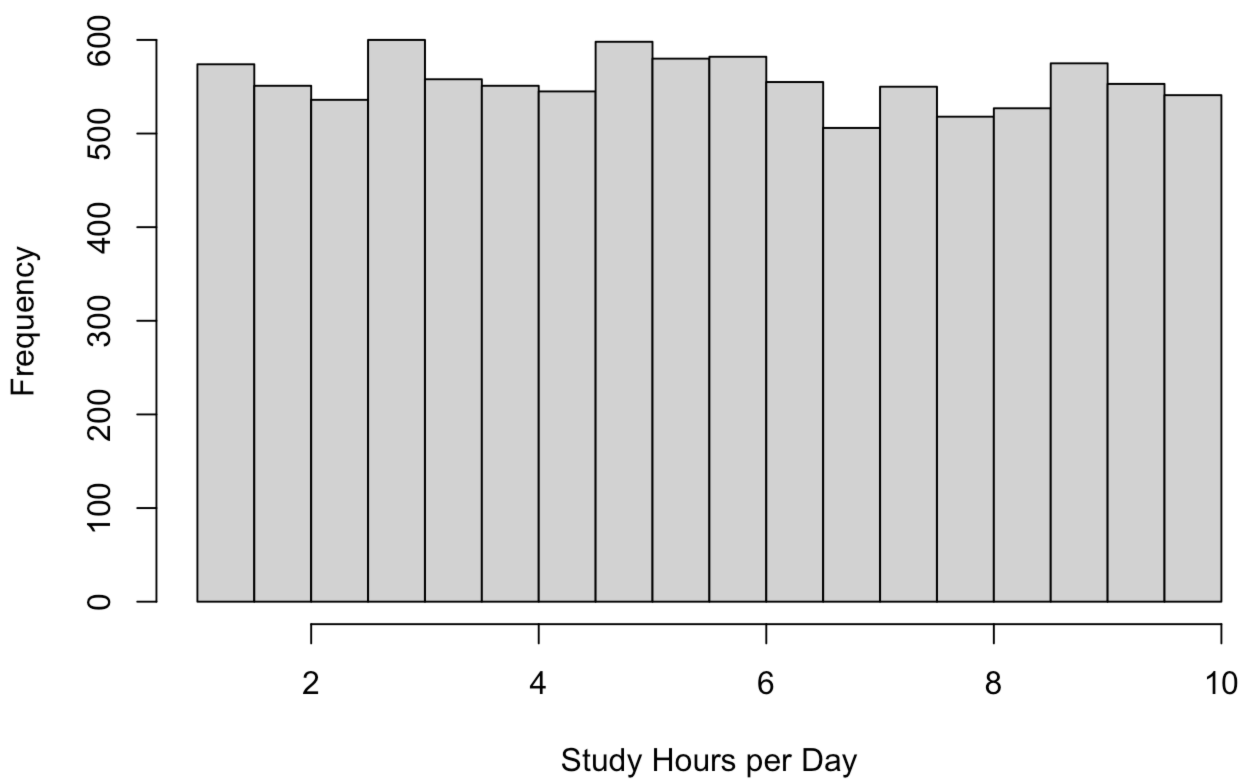
```
data$extra_classes <- factor(data$extra_classes)
summary(data$extra_classes)
```

```
##   No  Yes
## 4913 5087
```

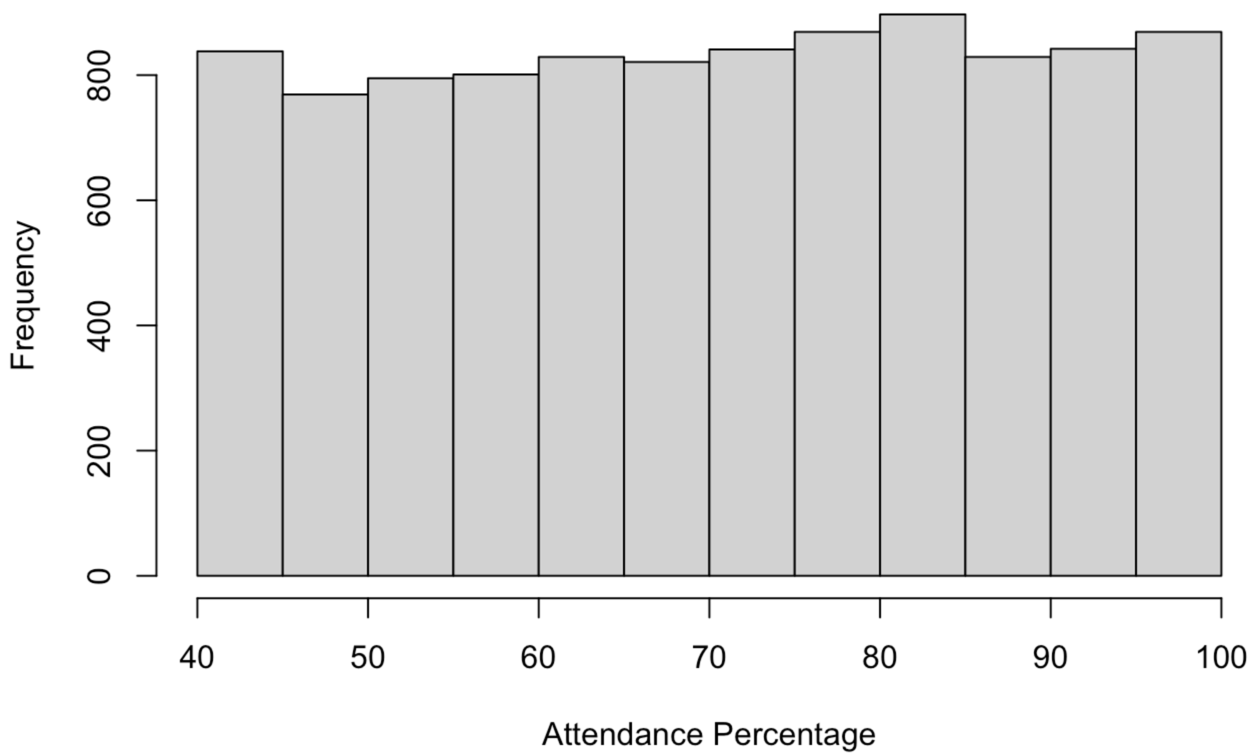
```
data$parent_education <- factor(data$parent_education)
summary(data$parent_education)
```

```
##    Bachelor High School      Master         PhD
##        2527        2528        2487        2458
```

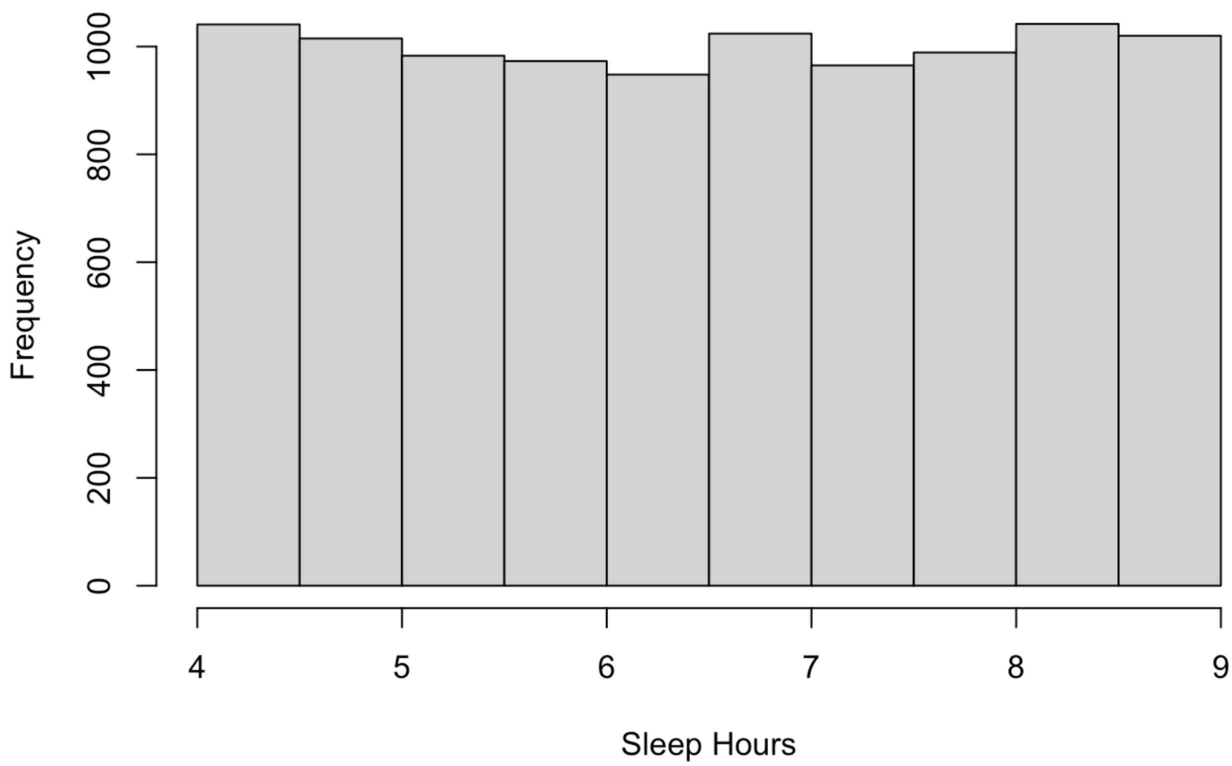
```
hist(data$study_hours_per_day, xlab="Study Hours per Day", main="")
```



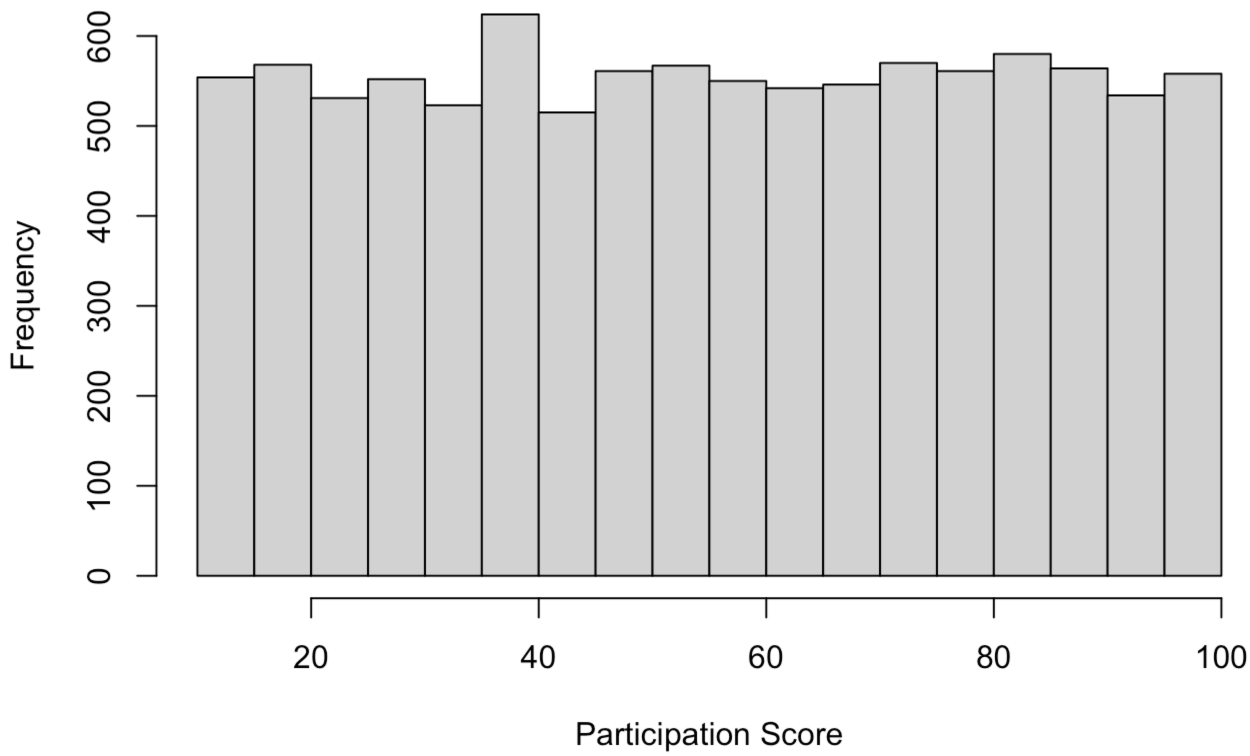

```
hist(data$attendance_percentage, xlab="Attendance Percentage", main="")
```



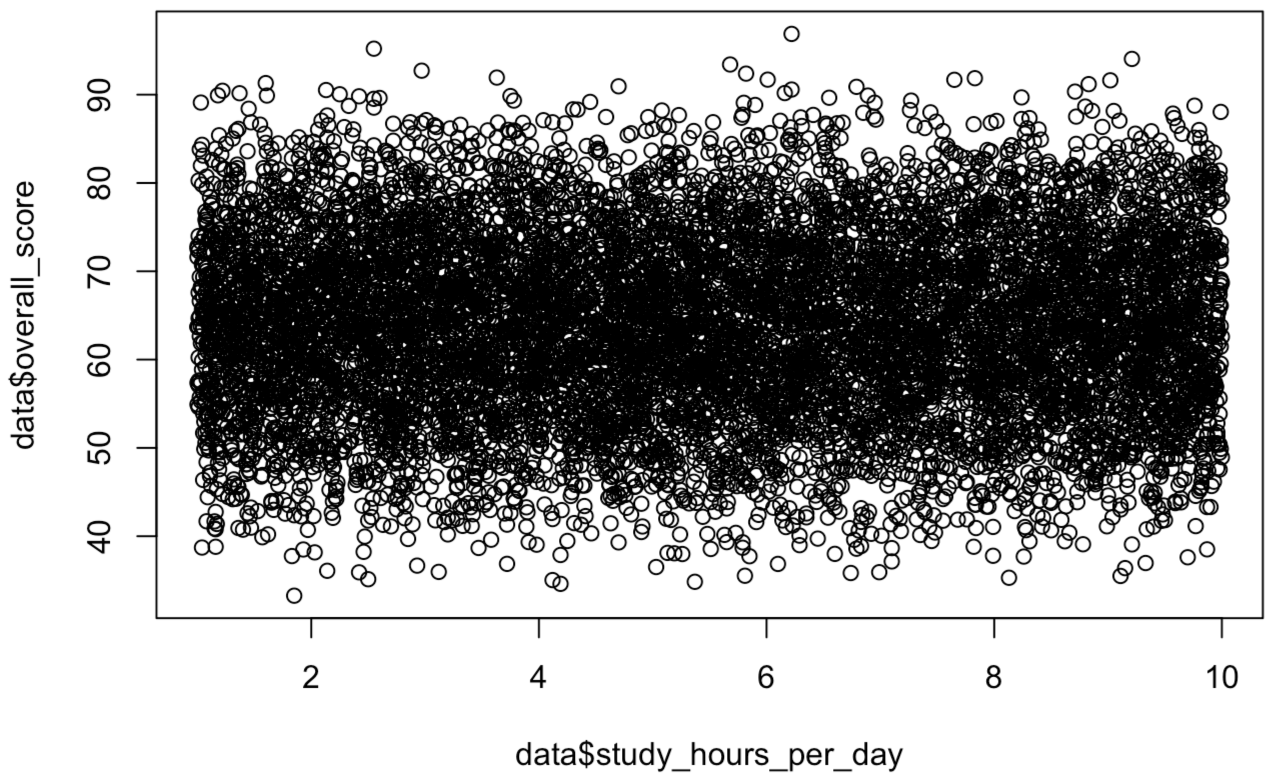

```
hist(data$sleep_hours, xlab="Sleep Hours", main="")
```




```
hist(data$participation_score, xlab="Participation Score", main="")
```




```
plot(data$overall_score ~ data$study_hours_per_day)
```



Task 2

```
mymodel<-lm(overall_score~., data=data)
summary(mymodel)
```

```
##
## Call:
## lm(formula = overall_score ~ ., data = data)
##
## Residuals:
##        Min         1Q     Median         3Q        Max
## -2.622e-12 -8.300e-15 -2.200e-15  3.900e-15  1.932e-11
##
## Coefficients:
##                               Estimate Std. Error    t value Pr(>|t|)
## (Intercept)                  3.789e-14  1.806e-14  2.099e+00   0.0359 *
## genderMale                   5.013e-15  3.952e-15  1.268e+00   0.2047
## study_hours_per_day         -4.536e-16  7.620e-16 -5.950e-01   0.5517
## attendance_percentage        1.000e-01  1.143e-16  8.747e+14   <2e-16 ***
## assignment_score             2.000e-01  9.859e-17  2.029e+15   <2e-16 ***
## midterm_score                2.500e-01  9.112e-17  2.744e+15   <2e-16 ***
## final_exam_score             3.500e-01  9.780e-17  3.579e+15   <2e-16 ***
## participation_score          1.000e-01  7.614e-17  1.313e+15   <2e-16 ***
## internet_accessYes           3.776e-15  3.958e-15  9.540e-01   0.3402
## extra_classesYes            -3.934e-15  3.956e-15 -9.940e-01   0.3201
## parent_educationHigh School  8.414e-16  5.561e-15  1.510e-01   0.8798
## parent_educationMaster       8.700e-15  5.584e-15  1.558e+00   0.1193
## parent_educationPhD          1.679e-16  5.600e-15  3.000e-02   0.9761
## sleep_hours                  1.710e-15  1.356e-15  1.261e+00   0.2073
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.976e-13 on 9986 degrees of freedom
## Multiple R-squared:      1,  Adjusted R-squared:      1
## F-statistic: 2.063e+30 on 13 and 9986 DF,  p-value: < 2.2e-16
```

```
data <- subset(data, select = -c(assignment_score, midterm_score, final_exam_score, part
icipation_score))
```

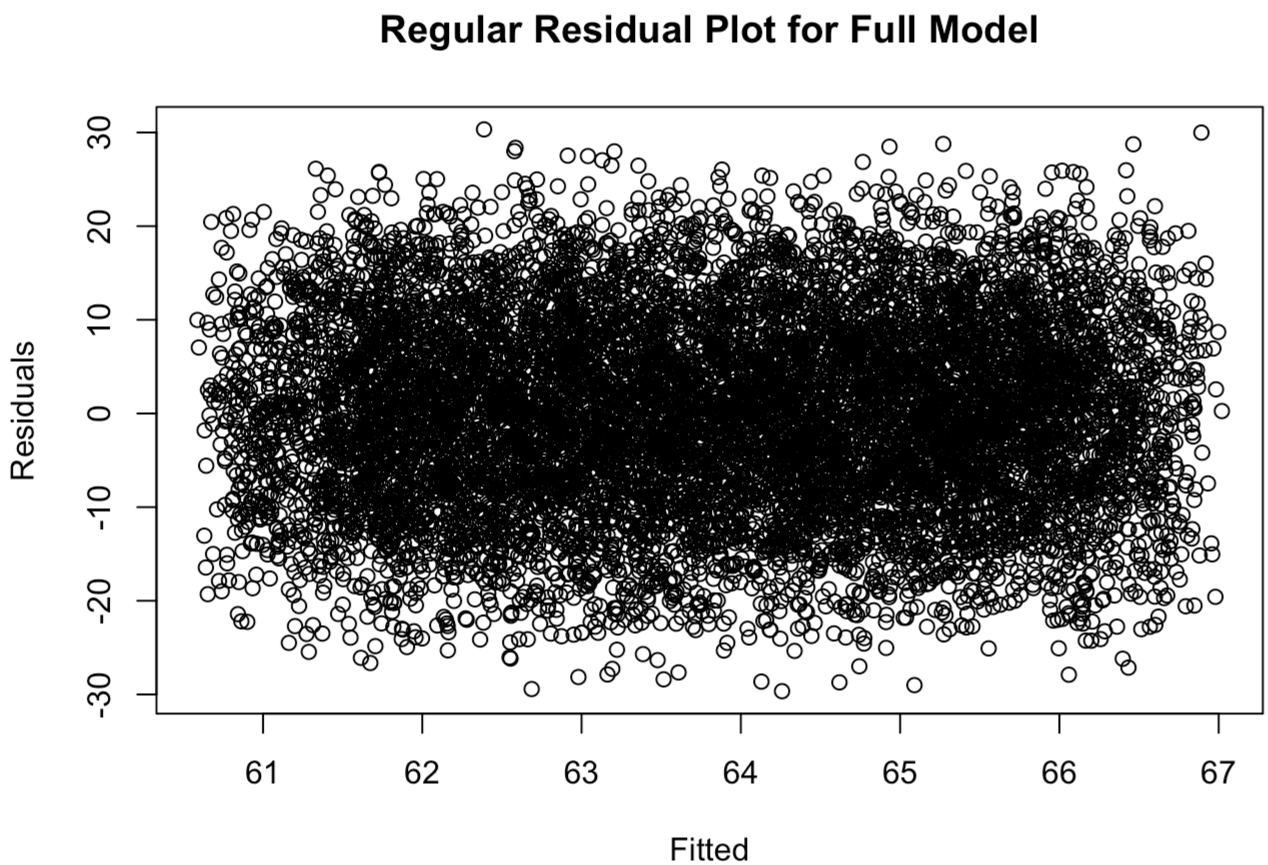
```
mymodel2<-lm(overall_score~.,data=data)
summary(mymodel2)
```

```
##
## Call:
## lm(formula = overall_score ~ ., data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -29.6517  -7.3093   0.0729   7.1936  30.3288
##
## Coefficients:
##                              Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 57.396903   0.708611  80.999  < 2e-16 ***
## genderMale                  -0.012119   0.202364  -0.060  0.95225
## study_hours_per_day         -0.004792   0.039011  -0.123  0.90223
## attendance_percentage        0.088141   0.005852  15.063  < 2e-16 ***
## internet_accessYes           0.640118   0.202500   3.161  0.00158 **
## extra_classesYes             0.148378   0.202508   0.733  0.46376
## parent_educationHigh School -0.283214   0.284657  -0.995  0.31979
## parent_educationMaster      -0.128598   0.285861  -0.450  0.65282
## parent_educationPhD         -0.335988   0.286702  -1.172  0.24126
## sleep_hours                  0.005474   0.069398   0.079  0.93713
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.12 on 9990 degrees of freedom
## Multiple R-squared:  0.02331,    Adjusted R-squared:  0.02243
## F-statistic: 26.49 on 9 and 9990 DF,  p-value: < 2.2e-16
```

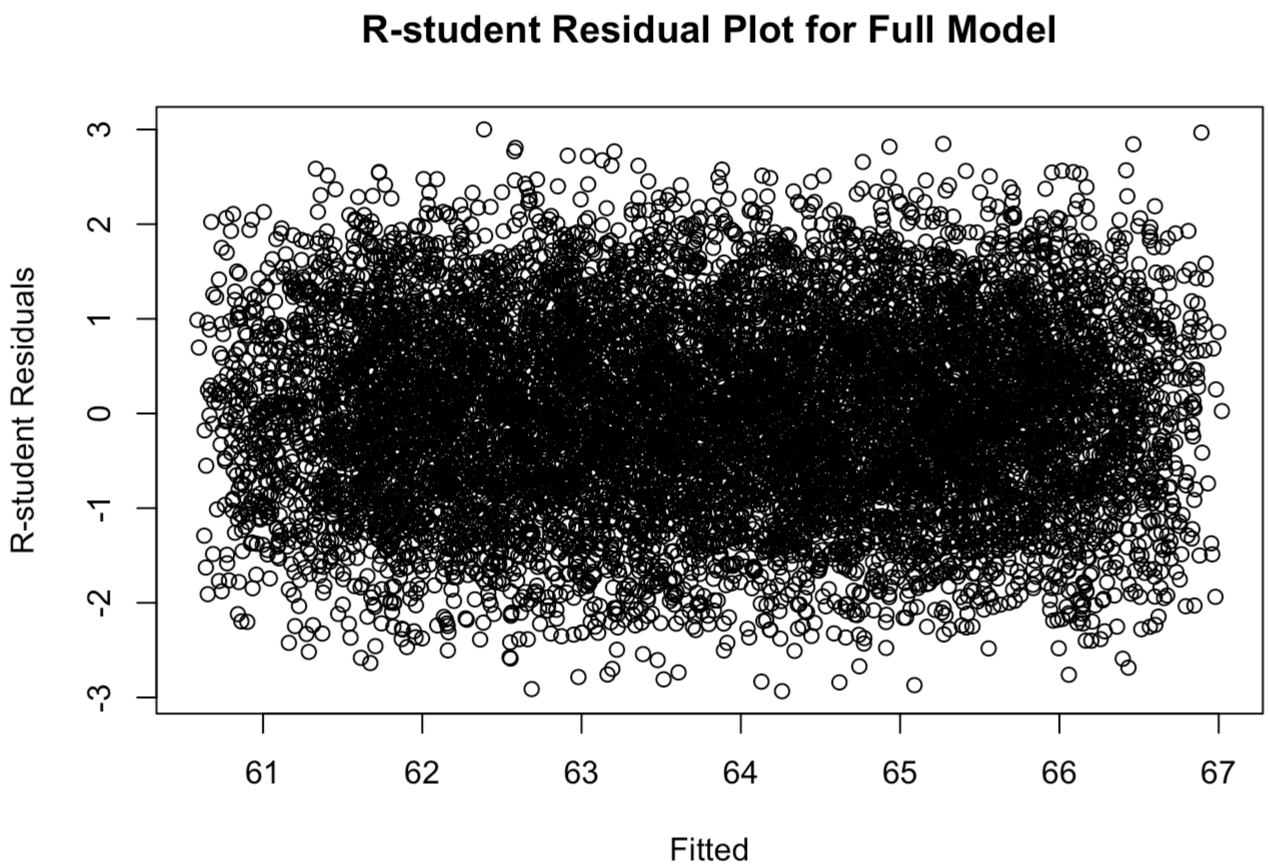
Task 3

a. Checking constant variance assumption for the errors

```
plot(fitted(mymodel2), residuals(mymodel2), xlab = "Fitted", ylab = "Residuals", main =
"Regular Residual Plot for Full Model")
```



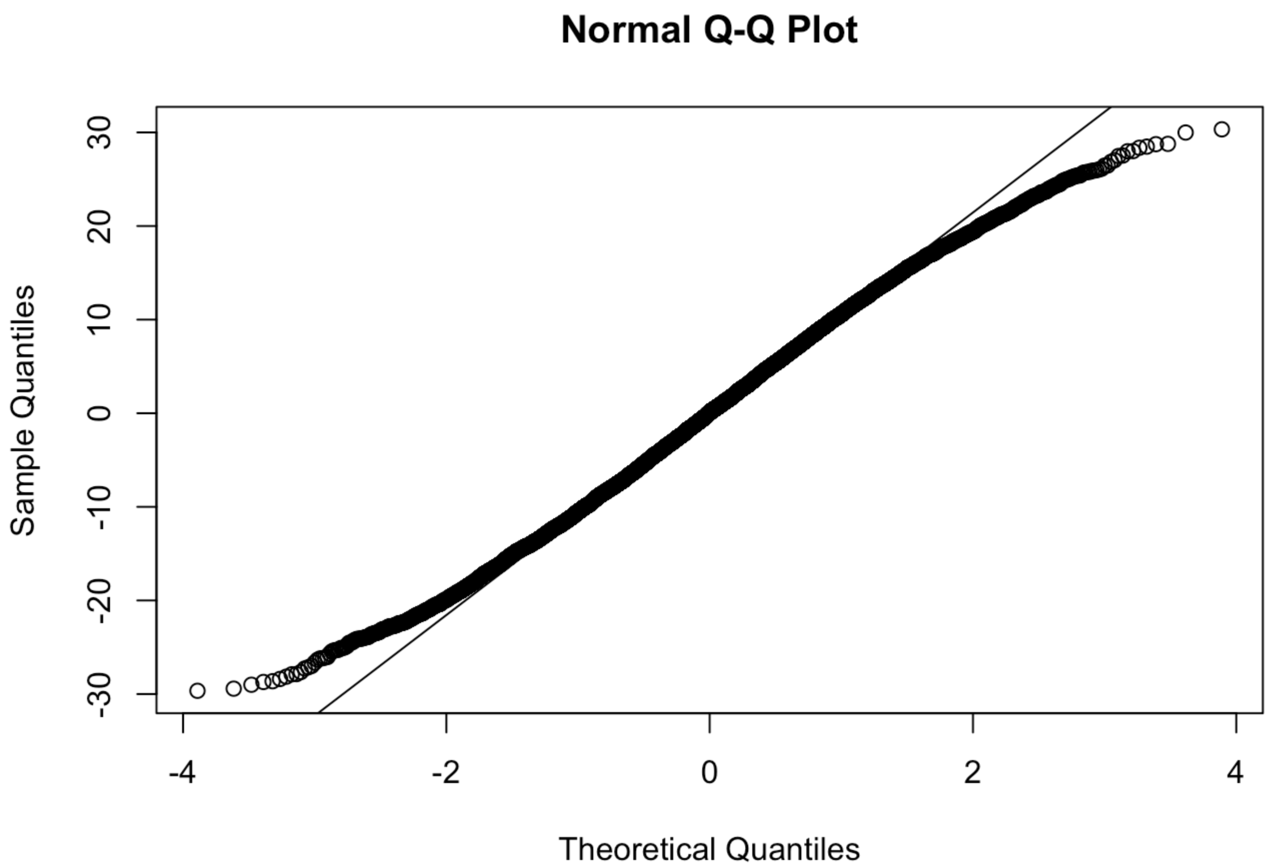

```
plot(fitted(mymodel2), rstudent(mymodel2), xlab = "Fitted", ylab = "R-student Residual
s", main = "R-student Residual Plot for Full Model")
```



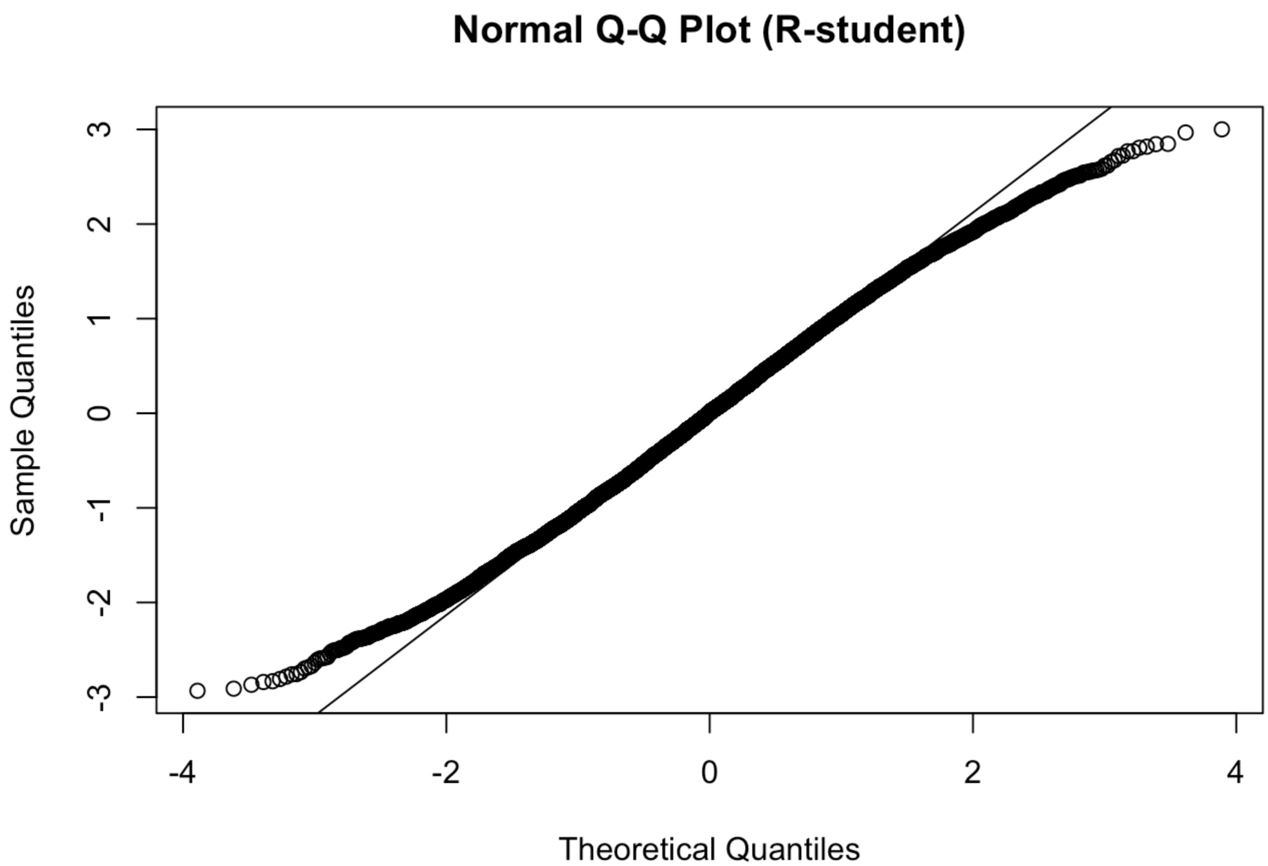
From both residual plots of the full model, the constant variance assumption does not seem to be violated. The points are spread out relatively evenly, following no specific pattern.

b. Checking normality assumption using qq-plots and normality tests

```
#qq plots
qqnorm(residuals(mymodel2), main = "Normal Q–Q Plot")
qqline(residuals(mymodel2))
```




```
qqnorm(rstudent(mymodel2), main = "Normal Q–Q Plot (R–student)")
qqline(rstudent(mymodel2))
```



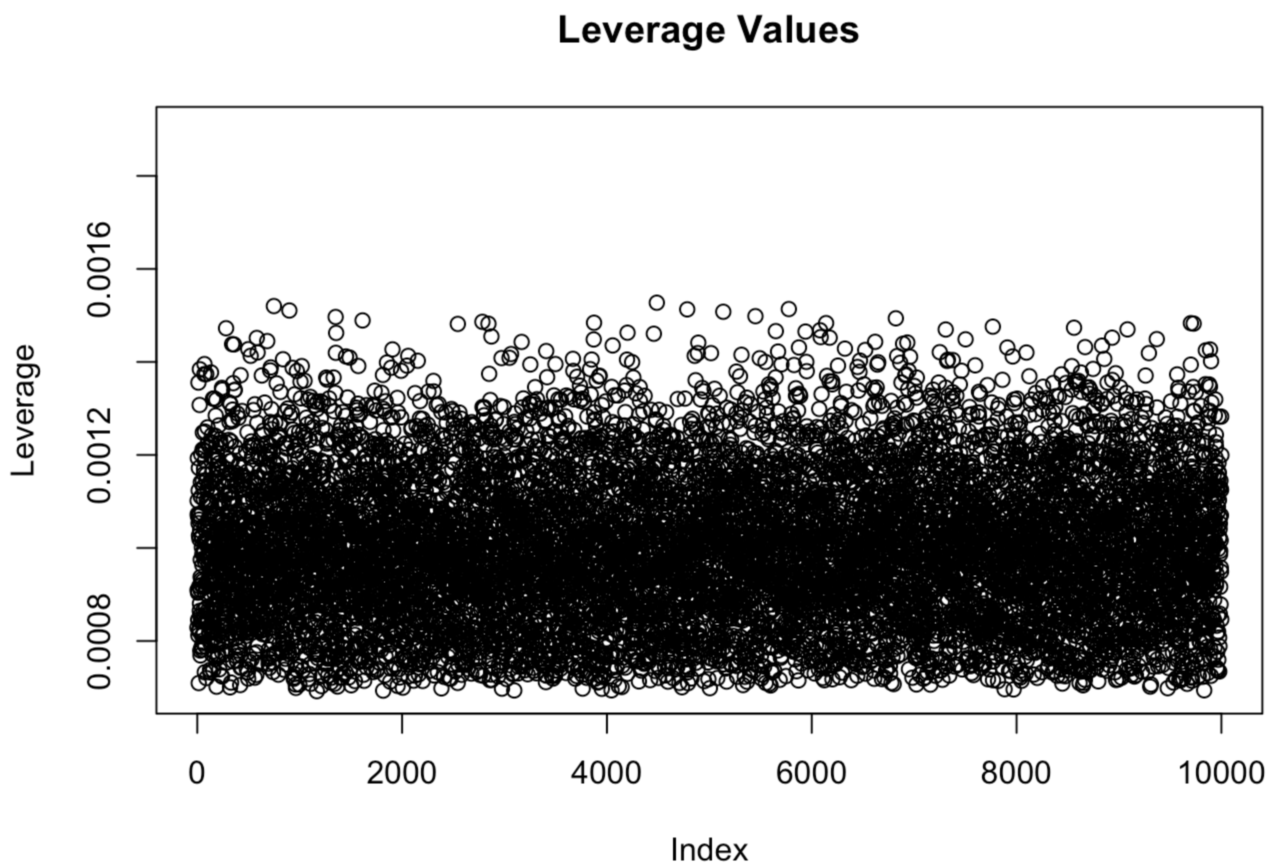

```
#normality tests --> cannot use shapiro test because n>5000
ks.test(residuals(mymodel2), "pnorm", mean = 0, sd =sd(residuals(mymodel2))) #kolmogorov
_Smirnov test
```

```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  residuals(mymodel2)
## D = 0.016142, p-value = 0.01091
## alternative hypothesis: two-sided
```

The QQ-plots suggest the normality assumption is not violated, however, the Kolmogorov Smirnov rejects normality with a p-value of 0.0323. But, with a large dataset, we know that even mild deviations from normality may be detected. However, these effects are mitigated by the large sample size. So, we can conclude that the data follows the normality assumption.

c. Checking for large leverage points

```
p = length(coef(mymodel2))
n = nrow(data)
h = lm.influence(mymodel2)$hat
plot(h, ylab = "Leverage", main = "Leverage Values", ylim = c(min(h), 0.0019))
abline(h = 2*p/n, lty =2,col = "red", lwd = 2)
```



There are no large leverage points. No point has a leverage above the threshold.

d. Check for outlier points in y

```
jack = rstudent(mymodel2)
jack[which.max(abs(jack))]
```

```
##     4668
## 3.000954
```

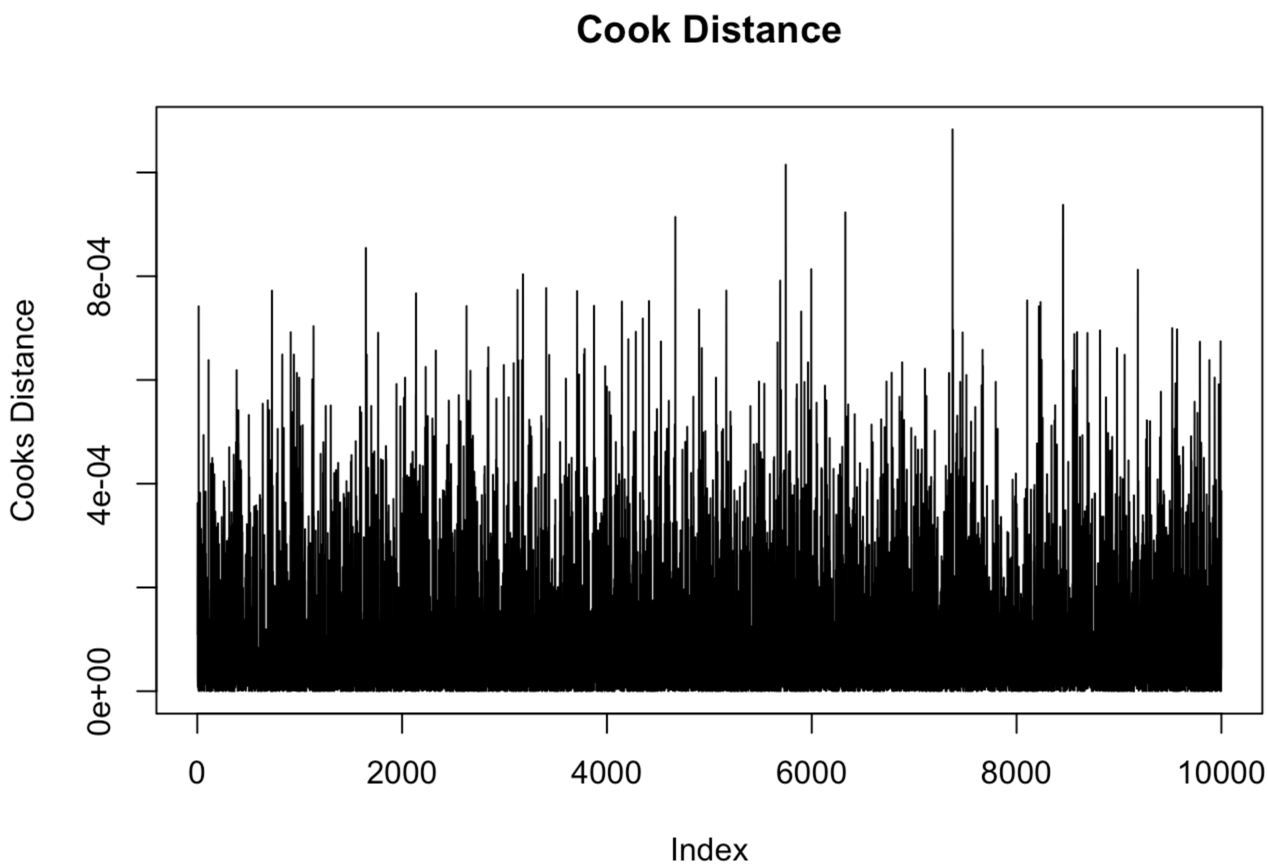
```
bonferroni_crit = qt(0.05/(n*2),n-p-1)
print(bonferroni_crit)
```

```
## [1] -4.567284
```

The largest R-student residual (absolute value) is below Bonferroni's critical value of abs(-4.567284) = 4.567284. Therefore, there are no outlier points in y.

e. Check for influential points

```
cook = cooks.distance(mymodel2)
plot(cook, ylab = "Cooks Distance", main = "Cook Distance", type = 'l')
```

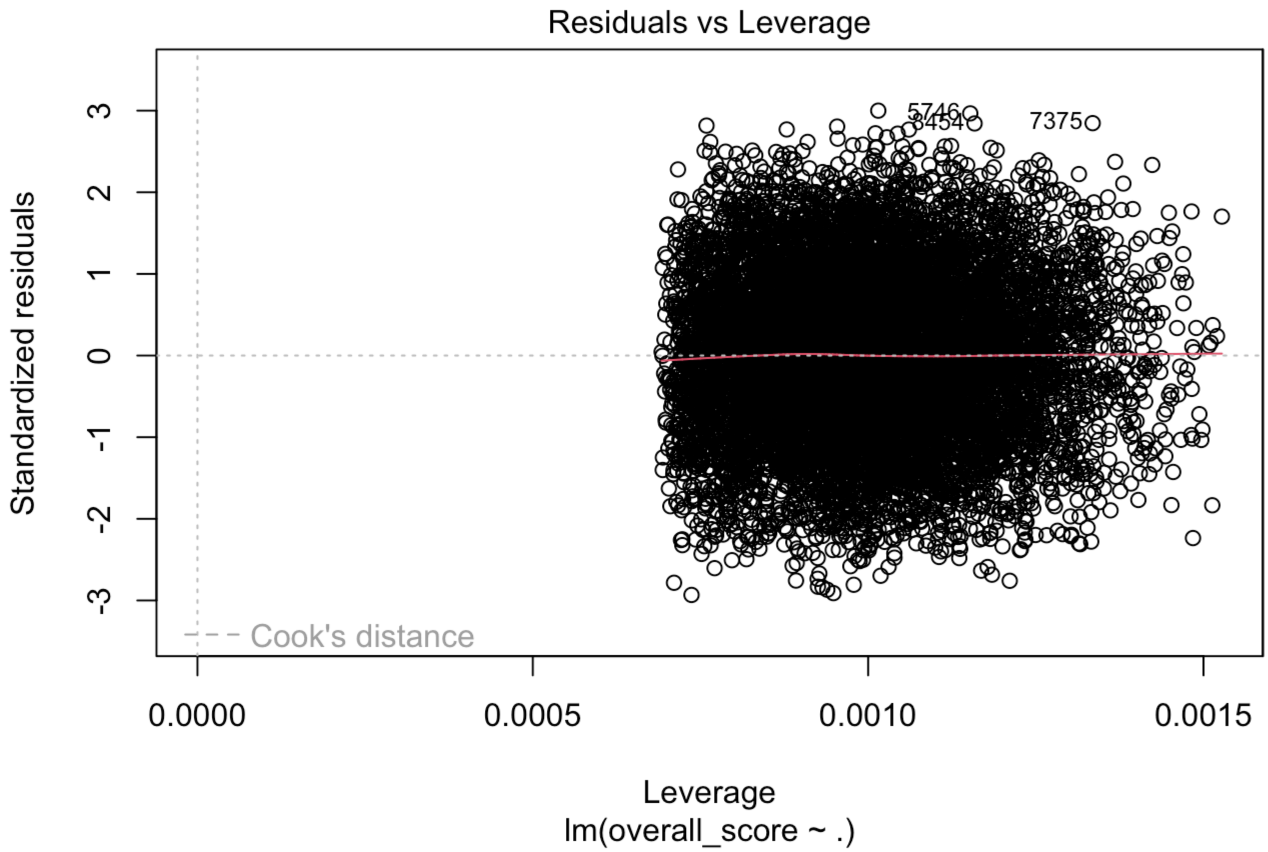



```
cook[which.max(cook)]
```

```
##        7375
## 0.001083028
```

```
  # width, height in inches

plot(mymodel2,which =5)
```

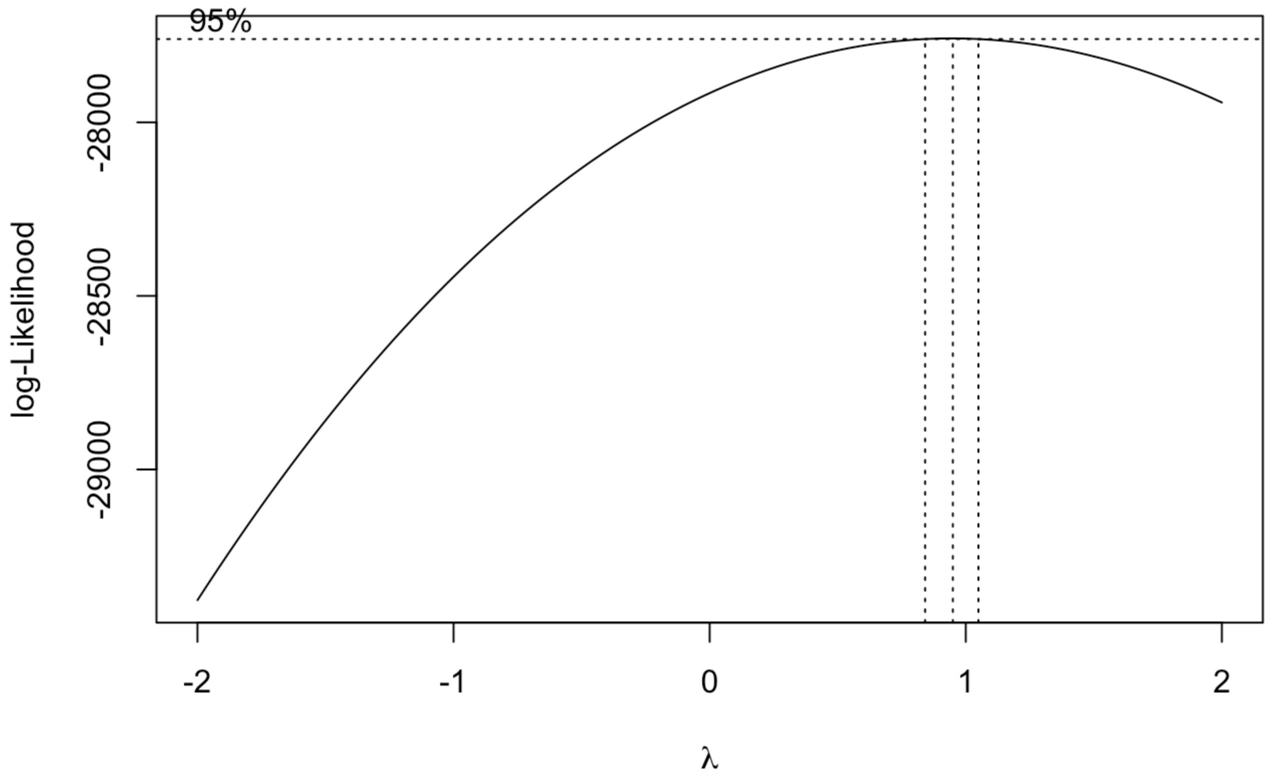


Cook’s Distance of all points < 0.5, suggesting there are no influential points. The max cook’s distance is 0.001083028.

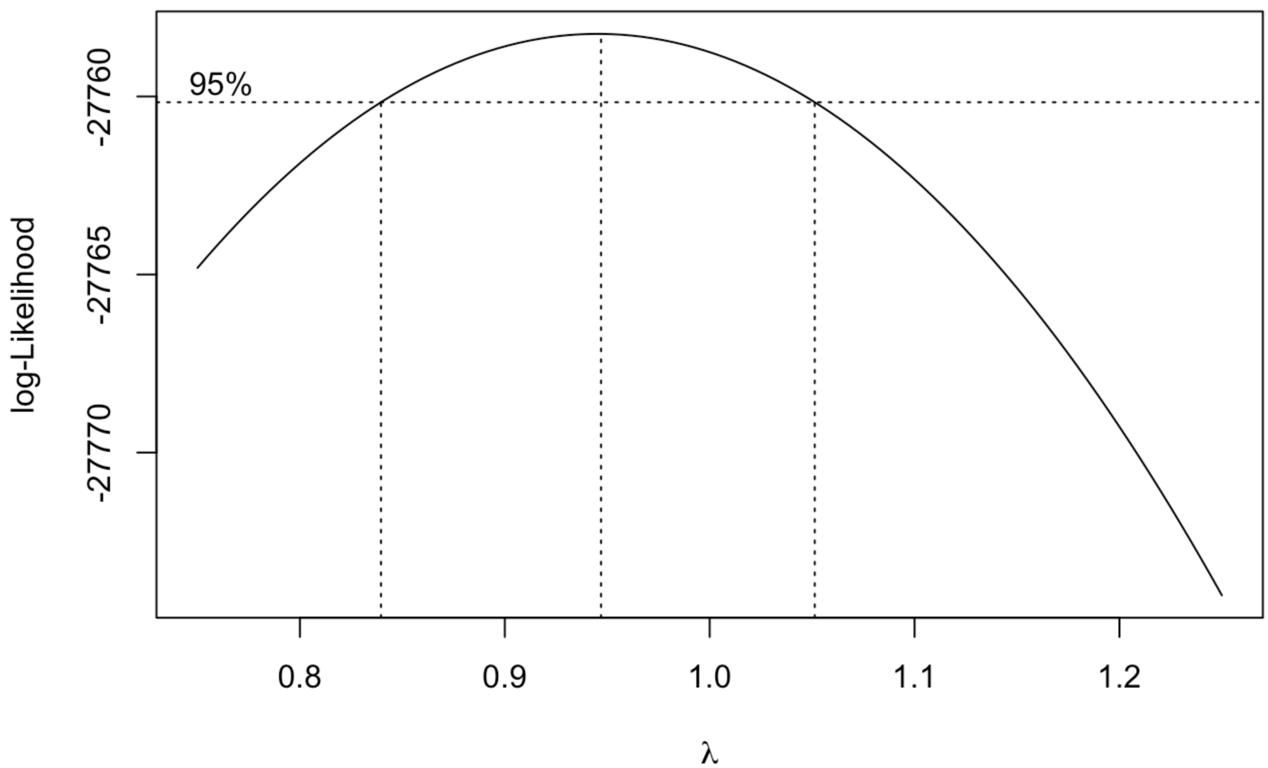
Task 4

Exploring possibility of applying a transformation. Although the errors are reasonably homoskedastic, we will try a transformation.

```
library(MASS)
boxcox(mymodel2, plotit = T)
```




```
boxcox(mymodel2, plotit=T, lambda = seq(0.75, 1.25, by = 0.05))
```



Because 1.0 falls the in the CI for lambda, there is no good reason to transform. So, the original model is appropriate.

Task 5

```
library(lmtest)
```

```
## Loading required package: zoo
```

```
## Warning: package 'zoo' was built under R version 4.5.2
```

```
##
## Attaching package: 'zoo'
```

```
## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric
```

```
library(nlme)
```

```
data_task5 <- subset(original_data, select = -c(grade, assignment_score, midterm_score,
final_exam_score, participation_score))
data_task5 <- data_task5[order(data_task5$student_id), ]

model_task5 <- lm(overall_score ~ ., data = data_task5)

dwtest(model_task5)
```

```
##
##  Durbin-Watson test
##
## data:  model_task5
## DW = 2.0089, p-value = 0.6677
## alternative hypothesis: true autocorrelation is greater than 0
```

Sorting the data by student ID does not reveal any autocorrelation. Therefore, applying a generalized least squares estimator is not necessary.

Task 6

```
bptest(mymodel2)
```

```
##
##  studentized Breusch-Pagan test
##
## data:  mymodel2
## BP = 2.2566, df = 9, p-value = 0.9867
```

The Breusch-Pagan test yields a p-value of 0.98. Since this is much greater than 0.05, we fail to reject the null hypothesis of homoskedasticity. This indicates that the variance of the errors is constant. However, for the sake of the assignment, we will create a WLS estimator anyway.

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following object is masked from 'package:nlme':
##
##     collapse
```

```
## The following object is masked from 'package:MASS':
##
##     select
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
data_task6 <- data %>%
  arrange(study_hours_per_day) %>%
  mutate(group = ntile(study_hours_per_day, n = 20))

group_stats <- data_task6 %>%
  group_by(group) %>%
  summarize(
    x_bar = mean(study_hours_per_day),
    s2_y = var(overall_score)
  )

variance_model <- lm(s2_y ~ x_bar, data = group_stats)

data_task6$predicted_var <- predict(variance_model, newdata = data.frame(x_bar = data_ta
sk6$study_hours_per_day))

data_task6$weights <- 1 / data_task6$predicted_var

wls_model <- lm(overall_score ~ ., data = data_task6, weights = weights)

summary(wls_model)
```

```
##
## Call:
## lm(formula = overall_score ~ ., data = data_task6, weights = weights)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -2.89960 -0.71431  0.00659  0.70723  2.97481
##
## Coefficients: (1 not defined because of singularities)
##                               Estimate Std. Error t value Pr(>|t|)
## (Intercept)                 -3.833e+03  2.180e+03  -1.758   0.0787 .
## genderMale                  -1.488e-02  2.024e-01  -0.074   0.9414
## study_hours_per_day          1.031e+01  5.952e+00   1.732   0.0832 .
## attendance_percentage        8.818e-02  5.852e-03  15.069   <2e-16 ***
## internet_accessYes           6.472e-01  2.025e-01   3.196   0.0014 **
## extra_classesYes             1.418e-01  2.026e-01   0.700   0.4840
## parent_educationHigh School -2.775e-01  2.847e-01  -0.975   0.3297
## parent_educationMaster      -1.245e-01  2.858e-01  -0.436   0.6632
## parent_educationPhD         -3.360e-01  2.867e-01  -1.172   0.2412
## sleep_hours                  4.093e-03  6.939e-02   0.059   0.9530
## group                        2.206e-01  3.463e-01   0.637   0.5241
## predicted_var                       NA         NA      NA       NA
## weights                      4.011e+05  2.247e+05   1.785   0.0743 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.9886 on 9988 degrees of freedom
## Multiple R-squared:  0.02362,    Adjusted R-squared:  0.02255
## F-statistic: 21.97 on 11 and 9988 DF,  p-value: < 2.2e-16
```

Applying a WLS estimator does not increase R-squared significantly (an increase of .00031).

Task 7

Pairwise correlations

```
round(cor(data[sapply(data, is.numeric)]), 3)
```

```
##                       study_hours_per_day attendance_percentage sleep_hours
## study_hours_per_day                 1.000                -0.014      -0.006
## attendance_percentage              -0.014                 1.000      -0.008
## sleep_hours                        -0.006                -0.008       1.000
## overall_score                      -0.004                 0.149      -0.001
##                       overall_score
## study_hours_per_day          -0.004
## attendance_percentage         0.149
## sleep_hours                  -0.001
## overall_score                 1.000
```

All correlations seem to be very close to 0 with the largest being 0.149 between attendance_percentage and overall_score. Solely based on the pairwise correlations, there is no evidence to suggest multicollinearity.

Variance Inflation Factors (VIF)

```
library("car")
```

```
## Warning: package 'car' was built under R version 4.5.2
```

```
## Loading required package: carData
```

```
## Warning: package 'carData' was built under R version 4.5.2
```

```
##
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':
##
##     recode
```

```
vif(mymodel2)
```

```
##                           GVIF Df GVIF^(1/(2*Df))
## gender                1.000512  1        1.000256
## study_hours_per_day   1.000788  1        1.000394
## attendance_percentage 1.000917  1        1.000458
## internet_access       1.001717  1        1.000858
## extra_classes         1.001644  1        1.000822
## parent_education      1.002436  3        1.000406
## sleep_hours           1.000758  1        1.000379
```

All values of GVIF are extremely close to 1 which again indicates that there is no multicollinearity.

Eigenvalues/Condition numbers

```
X <- model.matrix(mymodel2)[, -1]
e <- eigen(t(X) %*% X)
sqrt(e$values[1] / e$values)
```

```
## [1]   1.00000  23.92303  38.06552 141.98393 143.22628 146.04314 146.97318
## [8] 147.75177 282.72171
```

When looking at the condition numbers for the variables, they contradict the numbers derived from the pairwise correlations and the VIF. These numbers are likely inflated due to scaling differences and dummy variable encoding of categorical variables.

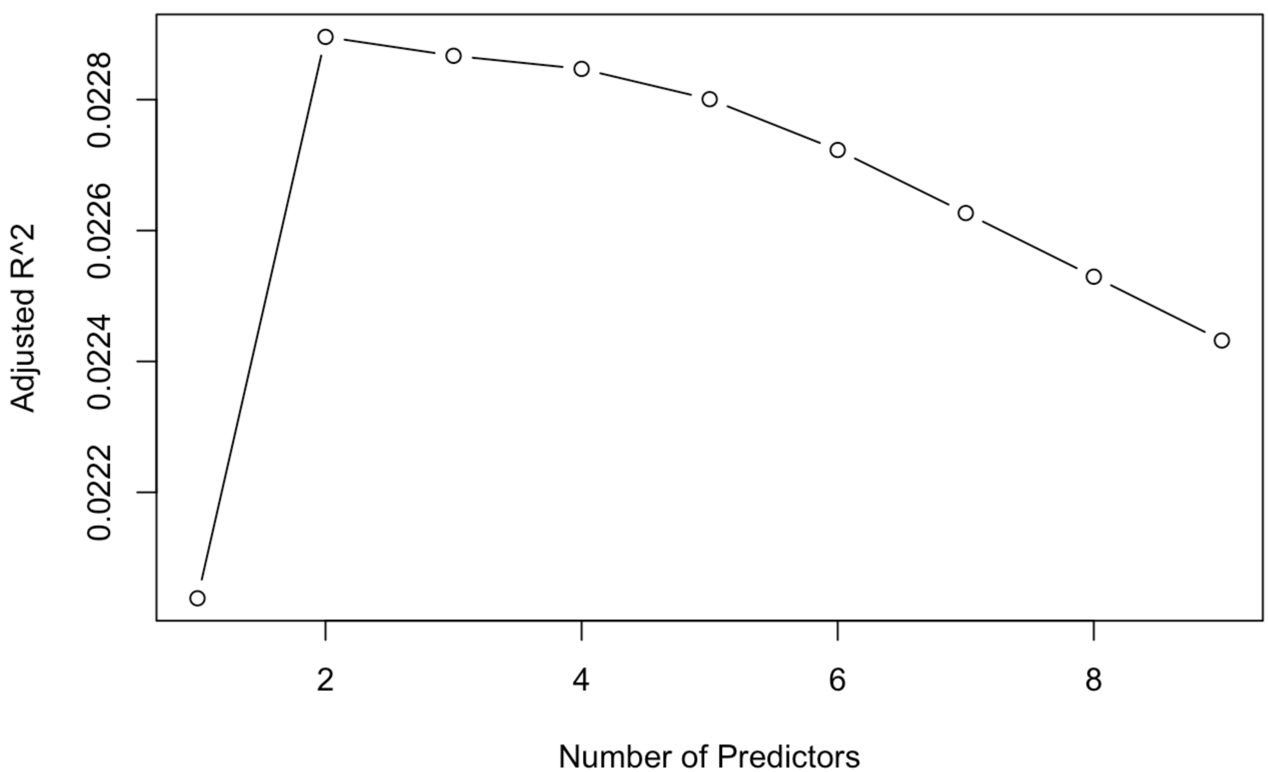
Task 8

Exhaustive Search

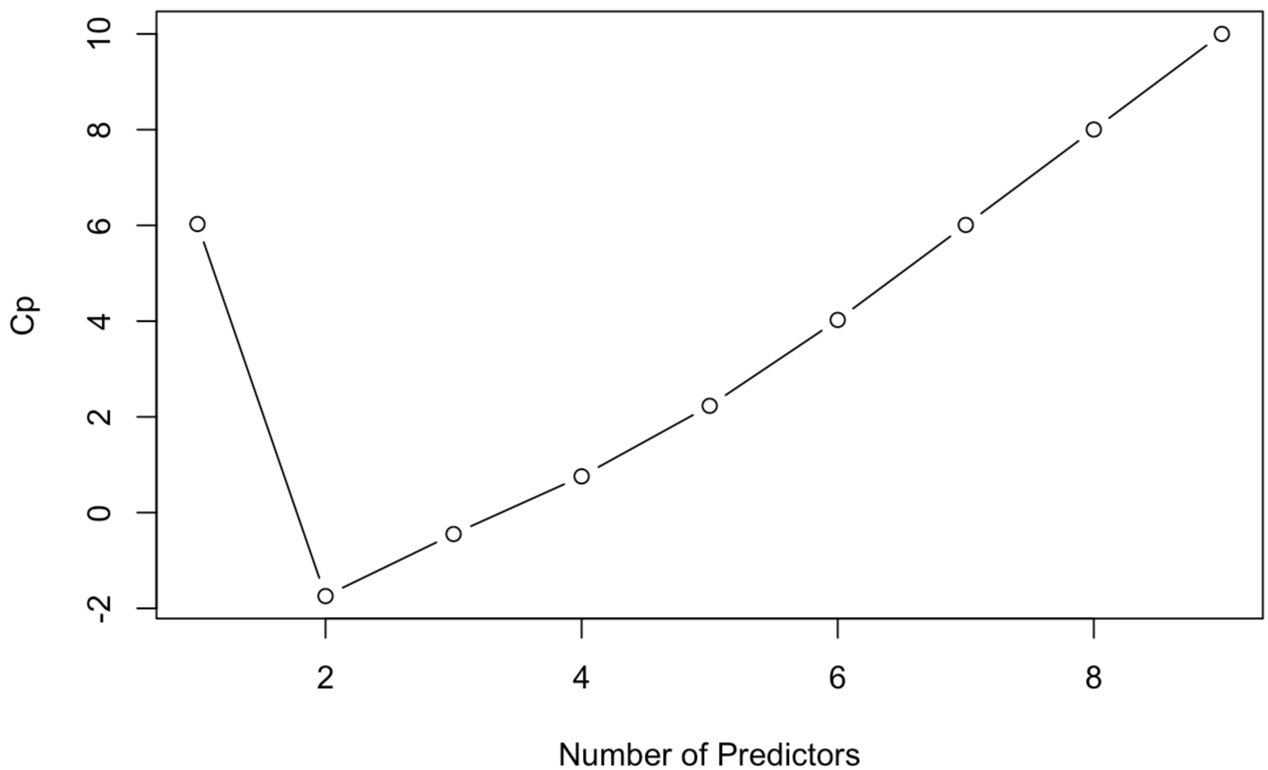
```
library(leaps)

regfit <- regsubsets(overall_score ~ ., data = data, nvmax = NULL)
reg_summary <- summary(regfit)

plot(reg_summary$adjr2, xlab = "Number of Predictors",
     ylab = "Adjusted R^2", type = "b")
```




```
plot(reg_summary$cp, xlab = "Number of Predictors",
     ylab = "Cp", type = "b")
```




```
which.max(reg_summary$adjr2)
```

```
## [1] 2
```

```
which.min(reg_summary$cp)
```

```
## [1] 2
```

```
coef(regfit, which.max(reg_summary$adjr2))
```

```
##          (Intercept) attendance_percentage    internet_accessYes
##          57.28679387            0.08823444            0.63253248
```

```
coef(regfit, which.min(reg_summary$cp))
```

```
##          (Intercept) attendance_percentage    internet_accessYes
##          57.28679387            0.08823444            0.63253248
```

Using the regsubsets() function, it was concluded through both the Adjusted R^2 plot and the Mallows' Cp plot that the optimal model size is two. The resulting fitted model that it selects is: overall_score = 57.287 + 0.0882 (attendance_percentage) + 0.6325 (internetaccess_Yes)

Task 9

Backward and forward model selection

```
full_mod <- mymodel2
null_mod <- lm(overall_score ~ 1, data = data)

back_mod <- step(full_mod, direction = "backward", trace = 0)
forw_mod <- step(null_mod,
                 scope = formula(full_mod),
                 direction = "forward",
                 trace = 0)

summary(back_mod)
```

```
##
## Call:
## lm(formula = overall_score ~ attendance_percentage + internet_access,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -29.5284  -7.3136   0.0764   7.2145  30.6069
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           57.286794   0.437830 130.843  < 2e-16 ***
## attendance_percentage  0.088234   0.005848  15.088  < 2e-16 ***
## internet_accessYes     0.632532   0.202294   3.127  0.00177 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.11 on 9997 degrees of freedom
## Multiple R-squared:  0.02309,    Adjusted R-squared:  0.0229
## F-statistic: 118.2 on 2 and 9997 DF,  p-value: < 2.2e-16
```

```
summary(forw_mod)
```

```
##
## Call:
## lm(formula = overall_score ~ attendance_percentage + internet_access,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -29.5284  -7.3136   0.0764   7.2145  30.6069
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           57.286794   0.437830 130.843  < 2e-16 ***
## attendance_percentage  0.088234   0.005848  15.088  < 2e-16 ***
## internet_accessYes     0.632532   0.202294   3.127  0.00177 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.11 on 9997 degrees of freedom
## Multiple R-squared:  0.02309,    Adjusted R-squared:  0.0229
## F-statistic: 118.2 on 2 and 9997 DF,  p-value: < 2.2e-16
```

```
formula(back_mod)
```

```
## overall_score ~ attendance_percentage + internet_access
```

```
formula(forw_mod)
```

```
## overall_score ~ attendance_percentage + internet_access
```

```
AIC(full_mod, back_mod, forw_mod)
```

```
##          df      AIC
## full_mod 11 74672.29
## back_mod  4 74660.55
## forw_mod  4 74660.55
```

The back and forw stepwise selection procedures both used the same reduced model. This model has a lower AIC (74660.55) when compared with the full model (74672.29), so this shows a better balance between model fit and complexity. Therefore, this reduced model is preferred and will be used for the further diagnostic analysis.

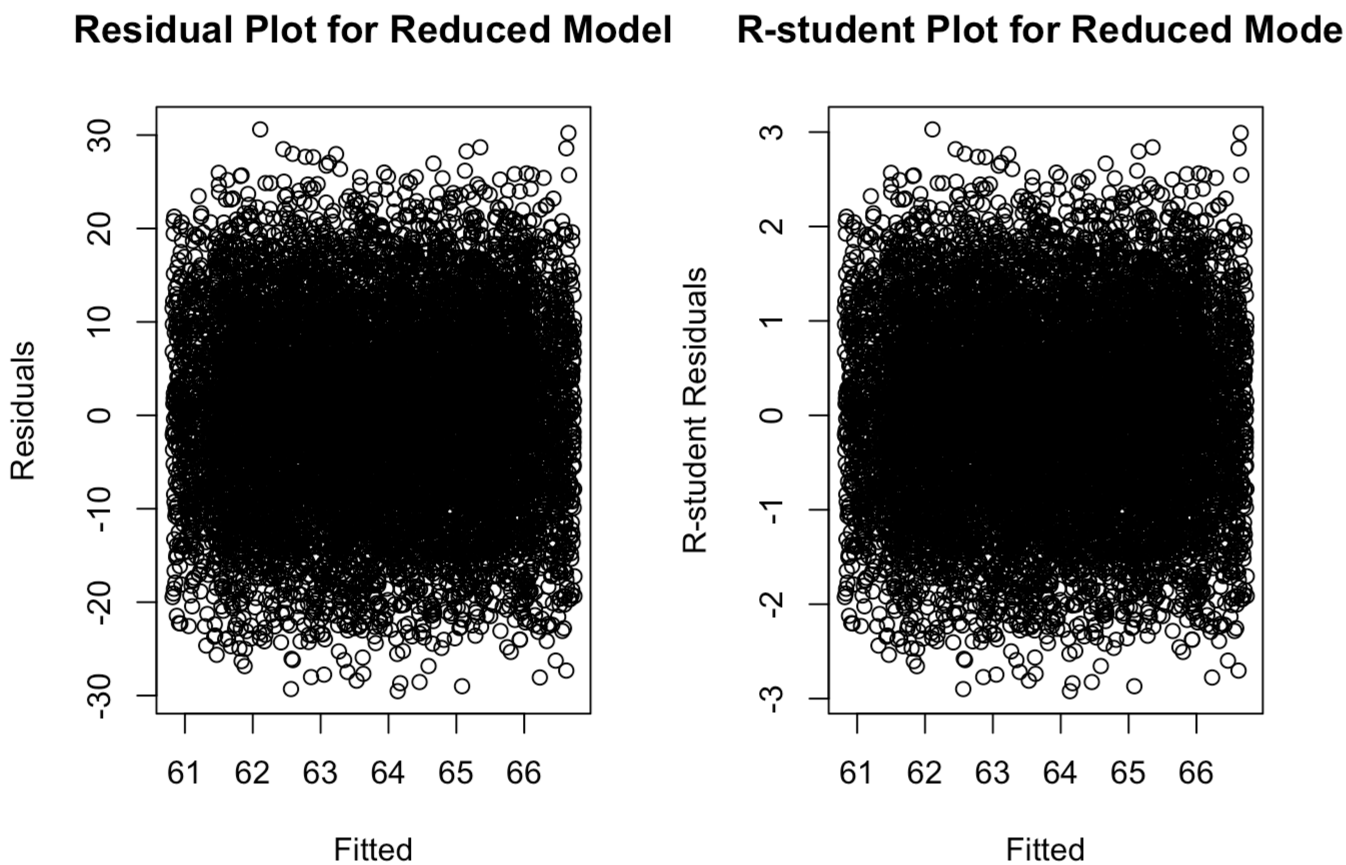
Task 10

a.

```
red_mod <- back_mod
summary(red_mod)
```

```
##
## Call:
## lm(formula = overall_score ~ attendance_percentage + internet_access,
##     data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -29.5284  -7.3136   0.0764   7.2145  30.6069
##
## Coefficients:
##                        Estimate Std. Error t value Pr(>|t|)
## (Intercept)           57.286794   0.437830 130.843  < 2e-16 ***
## attendance_percentage  0.088234   0.005848  15.088  < 2e-16 ***
## internet_accessYes     0.632532   0.202294   3.127  0.00177 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 10.11 on 9997 degrees of freedom
## Multiple R-squared:  0.02309,    Adjusted R-squared:  0.0229
## F-statistic: 118.2 on 2 and 9997 DF,  p-value: < 2.2e-16
```

```
par(mfrow = c(1,2))
plot(fitted(red_mod), residuals(red_mod),
     xlab = "Fitted", ylab = "Residuals",
     main = "Residual Plot for Reduced Model")
plot(fitted(red_mod), rstudent(red_mod),
     xlab = "Fitted", ylab = "R-student Residuals",
     main = "R-student Plot for Reduced Model")
```

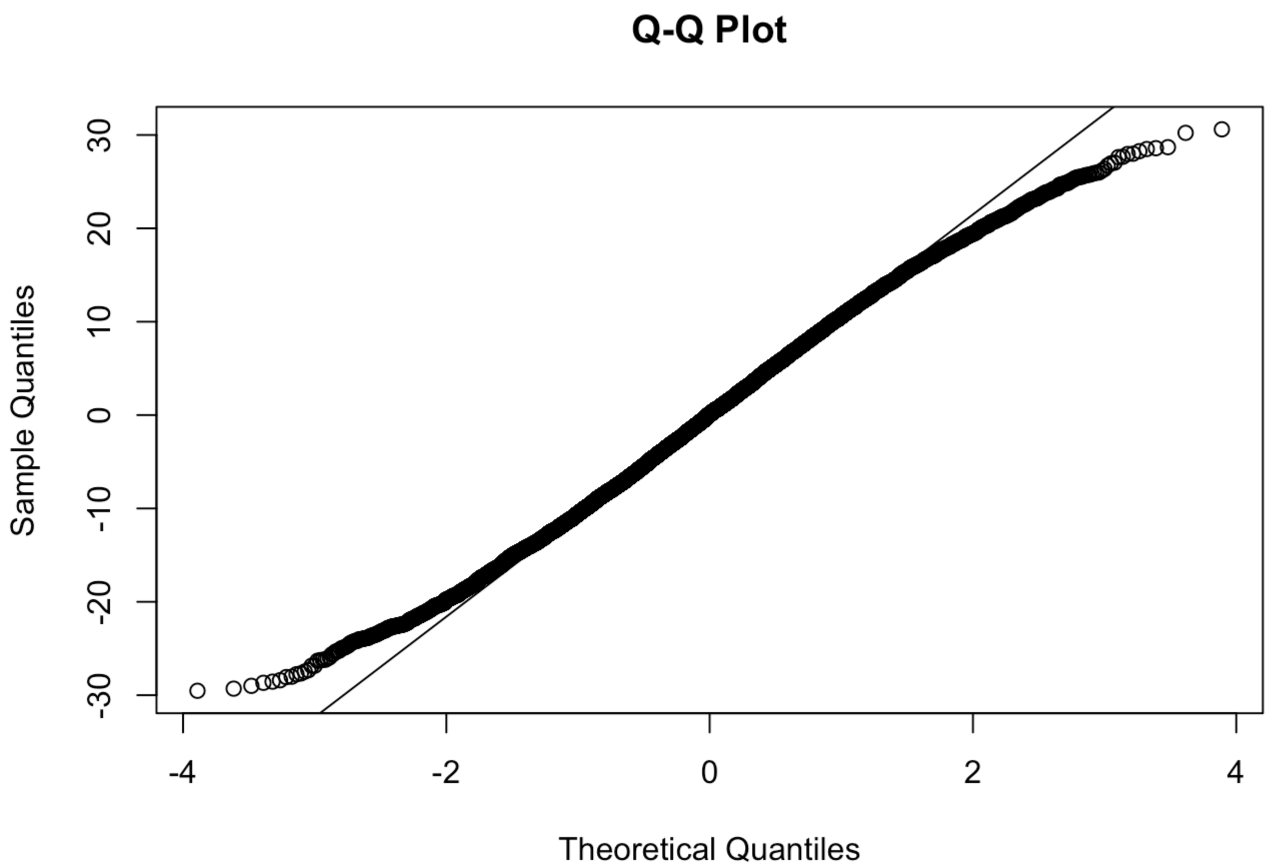



```
par(mfrow = c(1,1))
```

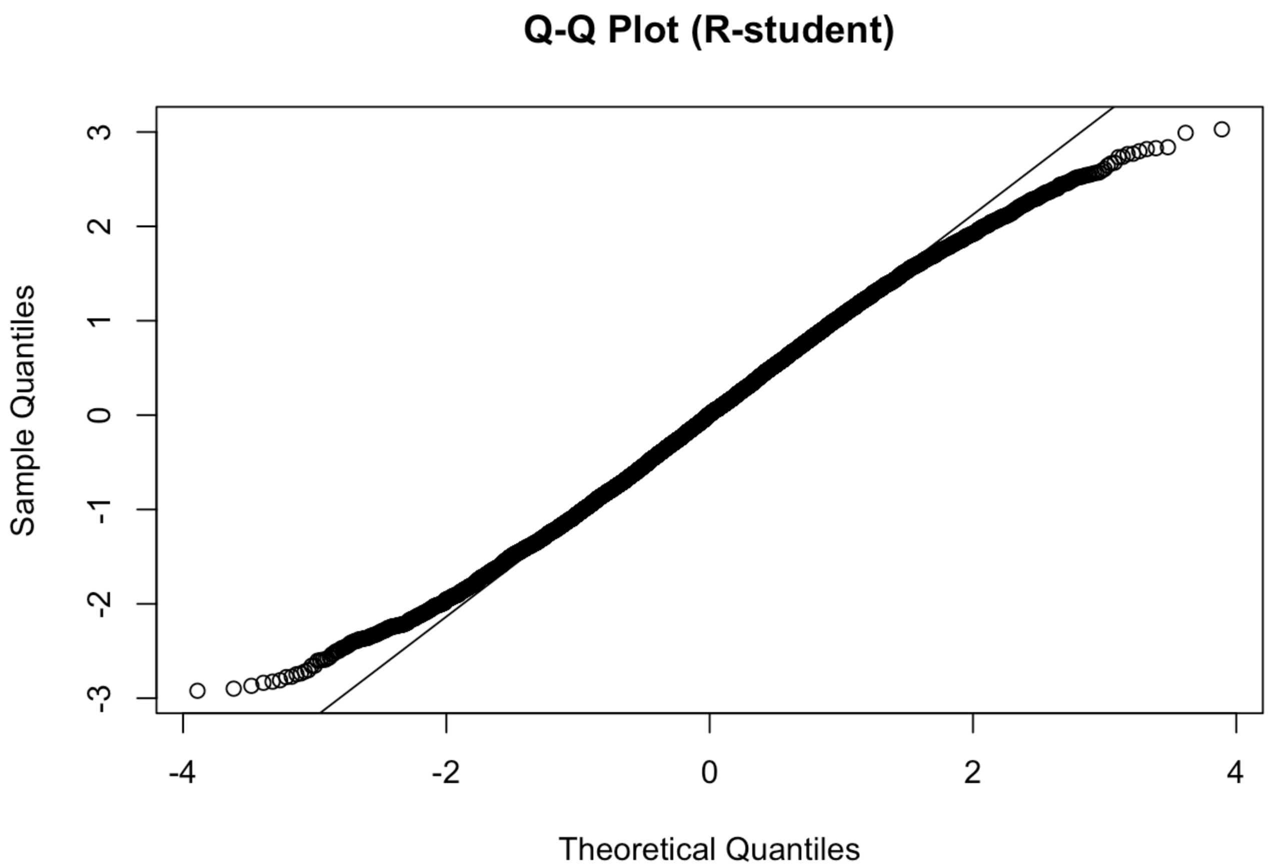
The residual and R-student plots don't show any clear pattern or change in spread, so the constant variance assumption seems to be satisfied.

b.

```
qqnorm(residuals(red_mod), main = "Q-Q Plot")
qqline(residuals(red_mod))
```




```
qqnorm(rstudent(red_mod), main = "Q-Q Plot (R-student)")
qqline(rstudent(red_mod))
```




```
ks.test(residuals(red_mod), "pnorm",
        mean = 0, sd = sd(residuals(red_mod)))
```

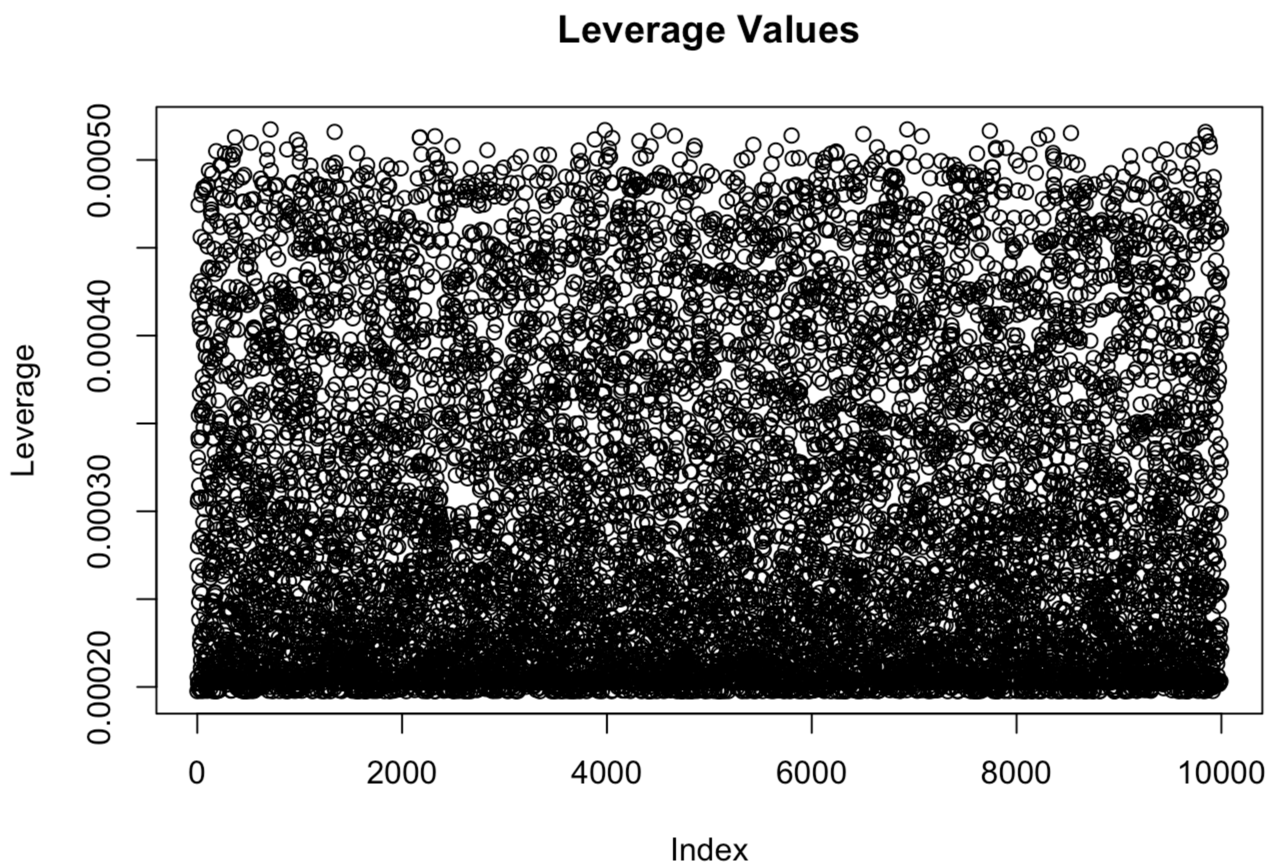
```
##
##  Asymptotic one-sample Kolmogorov-Smirnov test
##
## data:  residuals(red_mod)
## D = 0.016497, p-value = 0.008656
## alternative hypothesis: two-sided
```

The Q-Q plots show that there is slight deviations in the tails, and although the Kolmogorov-Smirnov test rejects normality (p-value = 0.008656), this is expected because of the large sample size we have. So the normality assumption is still considered to be reasonable.

c.

```
p_red <- length(coef(red_mod))
n_red <- nrow(data)
h_red <- lm.influence(red_mod)$hat

plot(h_red, ylab = "Leverage", main = "Leverage Values")
abline(h = 2 * p_red / n_red, lty = 2, col = "red", lwd = 2)
```




```
max(h_red)
```

```
## [1] 0.0005173938
```

```
2 * p_red / n_red
```

```
## [1] 6e-04
```

The maximum leverage value is 0.000517, which is less than the cutoff 2p/n=0.0006, indicating that there are no large leverage points.

d.

```
jack_red <- rstudent(red_mod)
jack_red[which.max(abs(jack_red))]
```

```
##     4668
## 3.028239
```

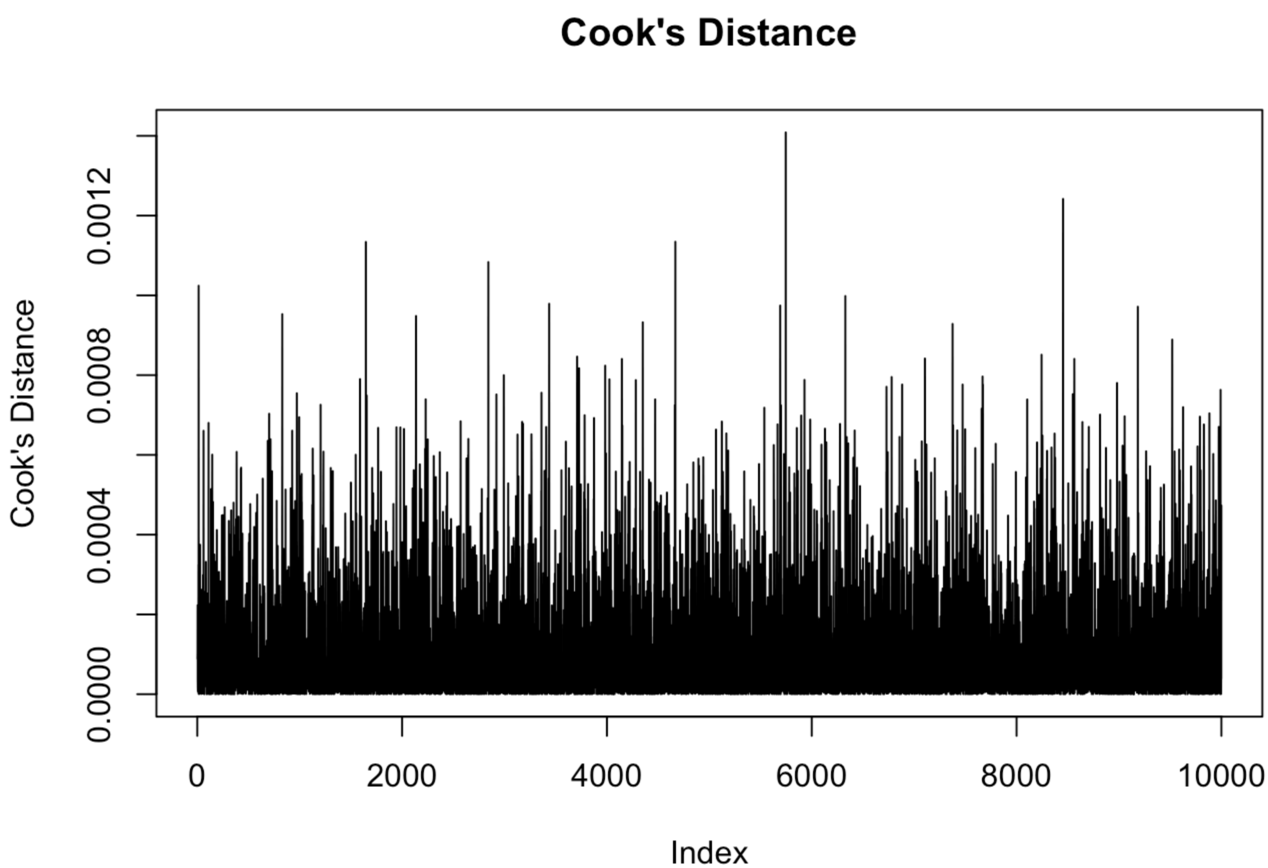
```
bonf_red <- qt(0.05 / (2 * n_red), n_red - p_red - 1)
bonf_red
```

```
## [1] -4.567282
```

The largest absolute R-student residual is 4.567, which is bigger than the Bonferroni critical value of 3.028. So, this indicates the presence of at least one outlier in the response.

e.

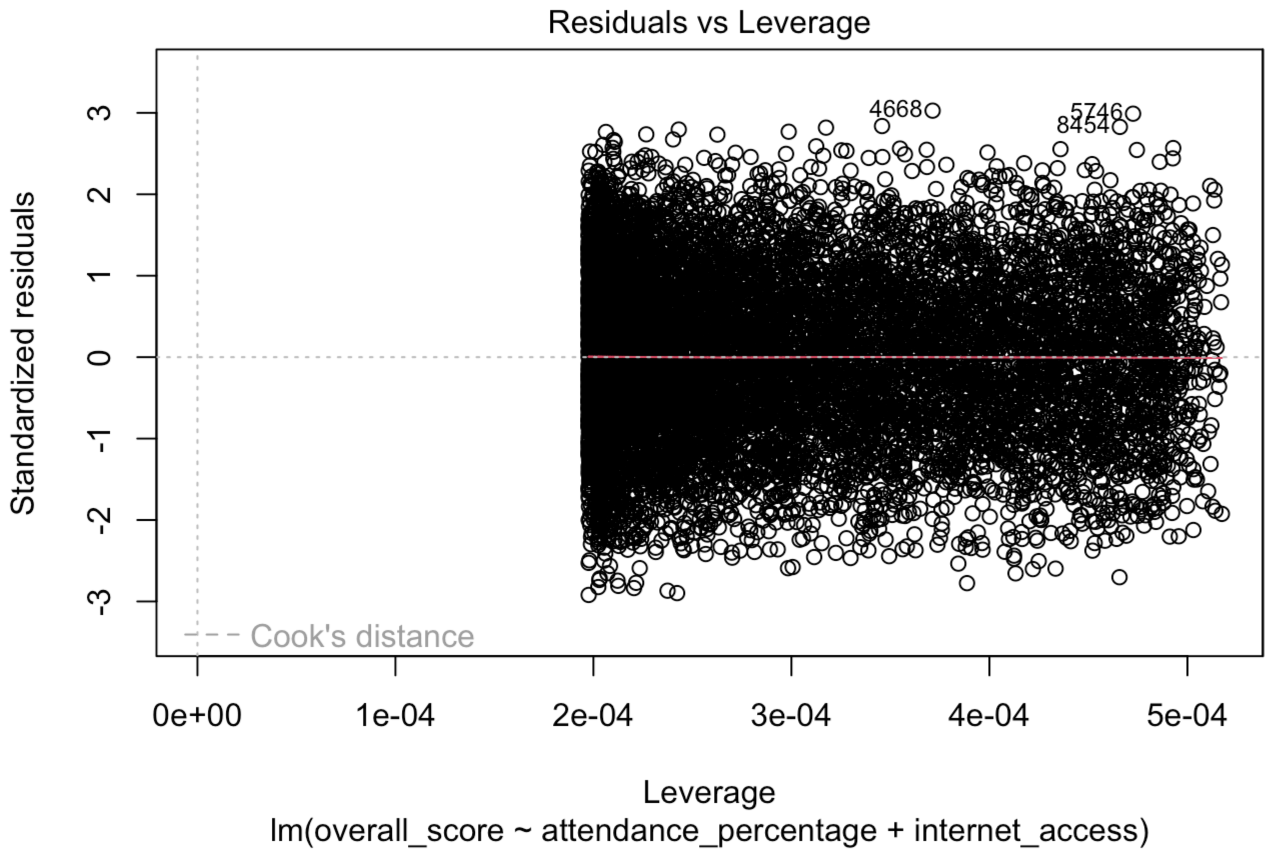
```
cook_red <- cooks.distance(red_mod)
plot(cook_red, ylab = "Cook's Distance", main = "Cook's Distance", type = "l")
```




```
cook_red[which.max(cook_red)]
```

```
##        5746
## 0.001408647
```

```
plot(red_mod, which = 5)
```



The max Cook's distance is very small at 0.00141, which indicates there are no influential observations.